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# STM32H5 scalable security

Tomas Team



# Introduction

- STM32H5 security is the last evolution of STM32 family
- It is based on STM32U5 and goes one step further
- The STM32U5 introduced the Secure AES with HUK and the keyed RDP
- The STM32H5 introduces
  - A new temporal isolation level
  - A secure storage coupled with temporal isolation
  - New product states
  - The Debug Authentication
  - An embedded secure boot called STiROT
  - Ready to use code example of two stage bootloader open sources
- You can use and combine those mechanisms to build your security !



# Agenda

1 Temporal isolation

5 STiROT

2 Secure storage

3 Product lifecycle

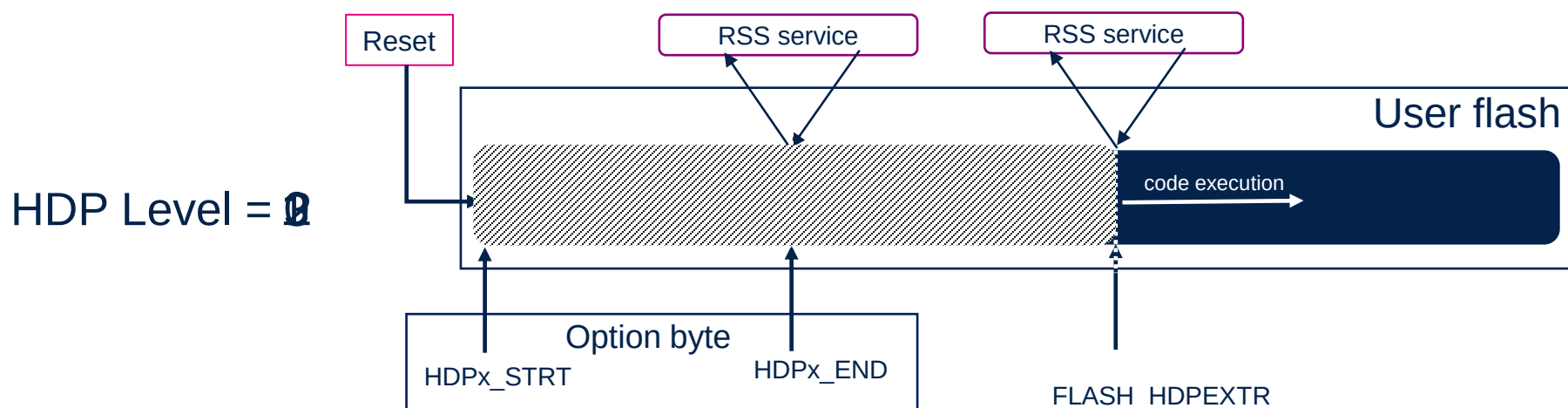
4 Secure boot path

# Temporal isolation



# What is temporal isolation ?

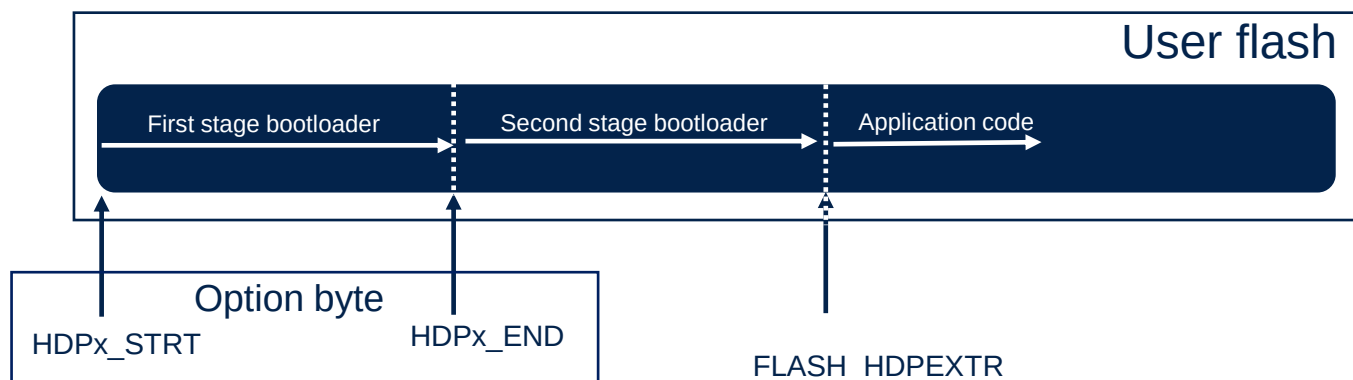
- Temporal isolation : protection that evolves during the code execution at boot
- STM32H5 has a temporal isolation named HDP
- Principle : 2 portions of user flash could be hidden until the next reset





# Temporal isolation on STM32H5

- 3 HDP Levels which allows to protect up to 2 user flash region.
- At boot all the user flash is available, increasing the HDP level allows to hide an associated region until the next boot.
- This has been designed to address 2 stage bootloader architecture



# Secure storage aka OBK



# Secure Storage

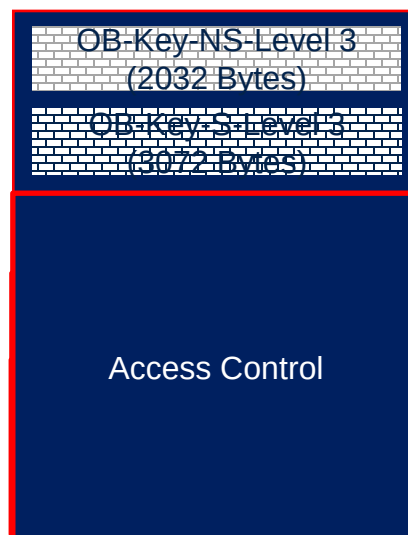
- Definition : Secure Storage is a specific area that allows to store data securely
  - data can be provisioned securely
  - data are isolated and can be accessed only by authorized part of the system
  - data can't be extracted from the system or re-used
- STM32H5 provide a secure storage named OBK storage
- Data stored are protected thanks to
  - Temporal isolation HDP
  - TrustZone / MPU isolation
  - Optionally : encryption mechanism based on a Hardware Unique Key and a monotonic counter



# Secure Storage

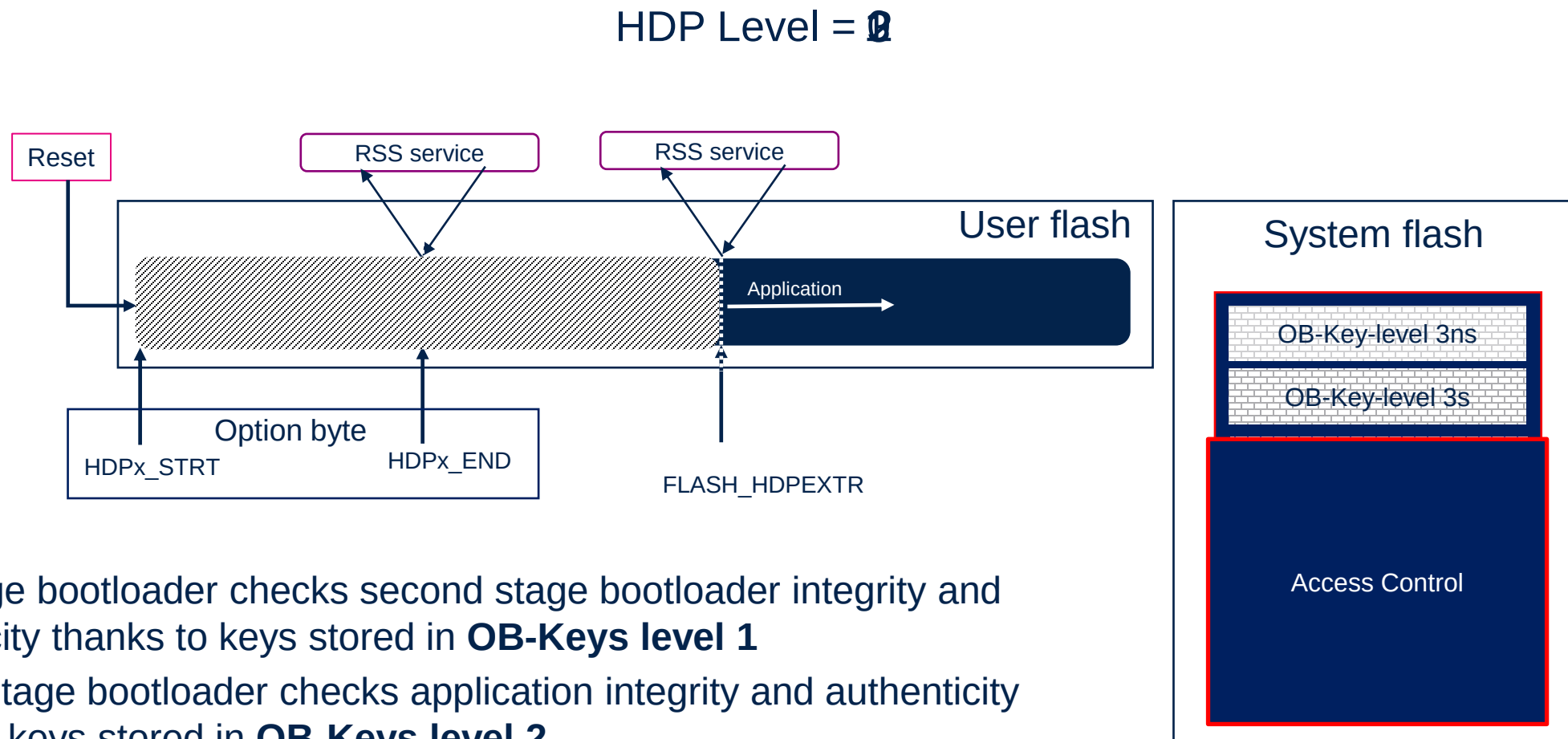
- OBK (option-byte keys) is a specific storage in system flash
- Total storage size is 8KB
- Split in 5 predefined areas corresponding to HDPL0, L1, L2, L3s, L3ns

HDPL = 0





# Temporal isolation and Secure storage



- First stage bootloader checks second stage bootloader integrity and authenticity thanks to keys stored in **OB-Keys level 1**
- Second stage bootloader checks application integrity and authenticity thanks to keys stored in **OB-Keys level 2**



# Provisioning of OBKeys

- OB-keys content can be provisioned when PRODUCT\_STATE is PROVISIONING
- A dedicated provisioning mechanism is implemented
- Programmer just needs to send a binary file containing the keys with a predefined format
- Tools provide easy way to build this binary file from xml



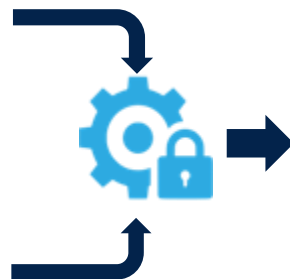


# STM32H5 Security OB-Key Provisioning



data.bin  
( Key1 )

Ob-key\_config.xml  
( Dest address  
Encryption )



OBKey\_config.obk

( Key1  
Dest address  
Encryption )

Prg

Provisioning

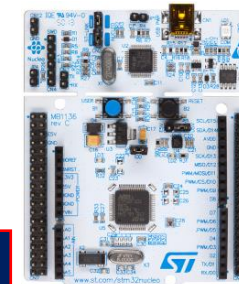
OB-Key-NS-Level 3

OB-Key-S-Level 3

OB-Key-Level 2

OB-Key-Level 1

OB-Key-level0



# Life cycle



# New Product States

- These were introduced earlier in the workshop
- Main additions:
  - Provisioning state: state dedicated to provision the secure storage
  - Closed state and Lock state : only difference is inability to connect the DA in lock state
  - Debug opening sub states
- Regression principle is the same as on other STM32



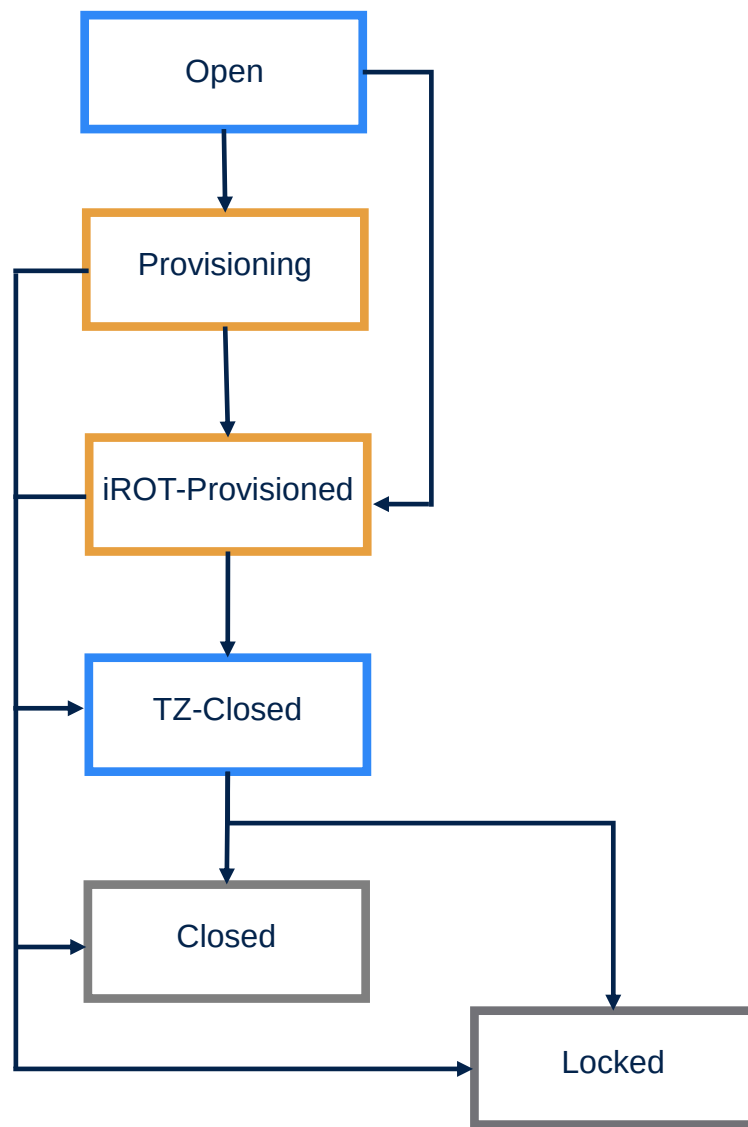
# STM32H5 Product Life Cycle

- Possible states :

|                  |                                                                |
|------------------|----------------------------------------------------------------|
| Open             | State fully open (default)                                     |
| Provisioning     | State to provision OB keys                                     |
| iROT-Provisioned | iRot has been provisioned                                      |
| TZ-Closed        | Only non secure is open. State only available if TZ is enabled |
| Closed           | Final state but with controlled debug /regression capabilities |
| Locked           | Final state without any reconnection possibility               |



# STM32H5 Product Life Cycle



} RDP level 0

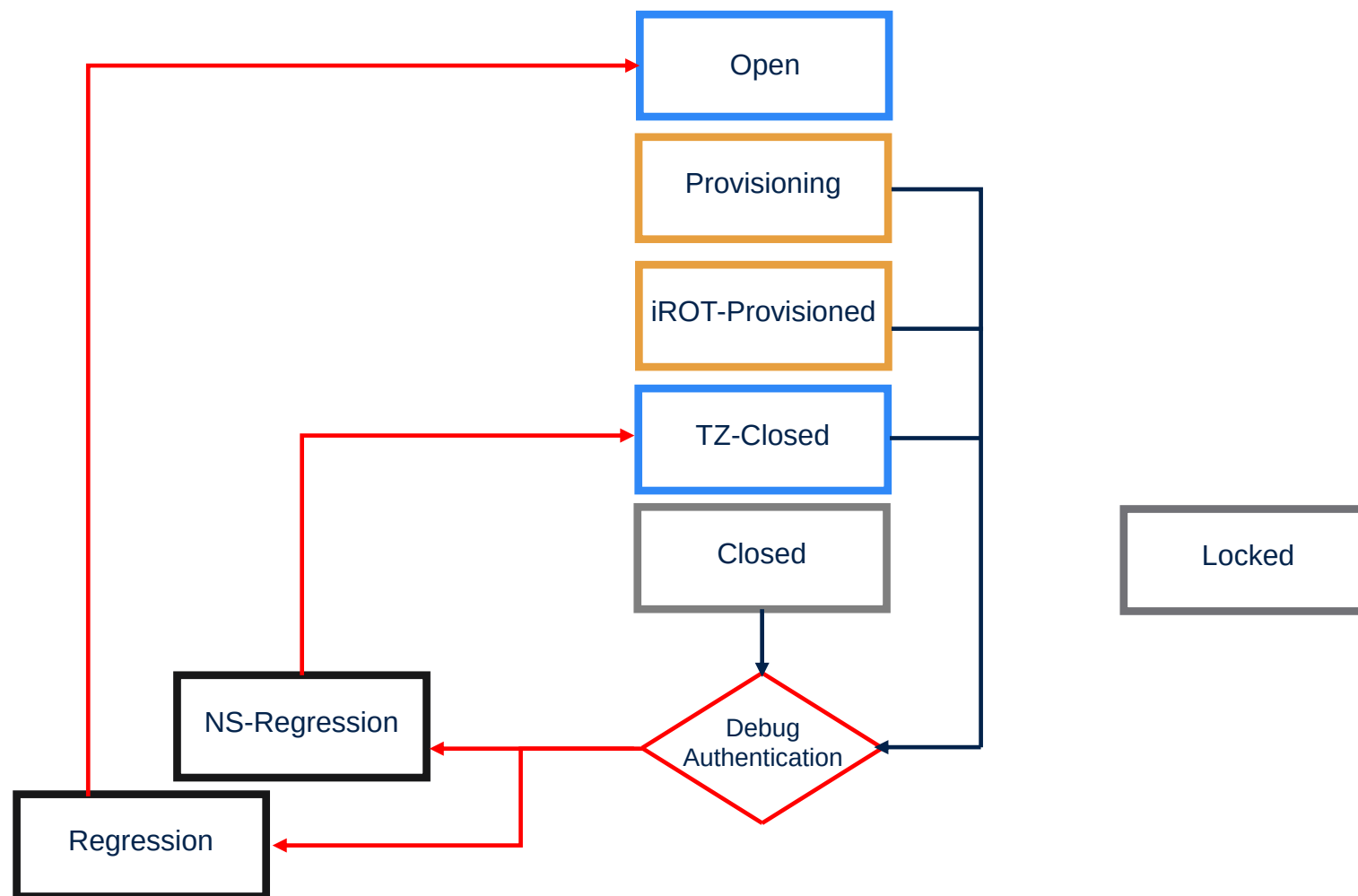
} RDP level 0.5

} Keyed RDP level 2

} RDP level 2



# STM32H5 Product Life Cycle – Regression





# STM32H5 Product Life Cycle

NS-Regression

NS flash code is erased  
OBK HDPL3NS are erased  
EPOCH\_NS incremented  
Similar to RDP1 to RDP0.5 regression

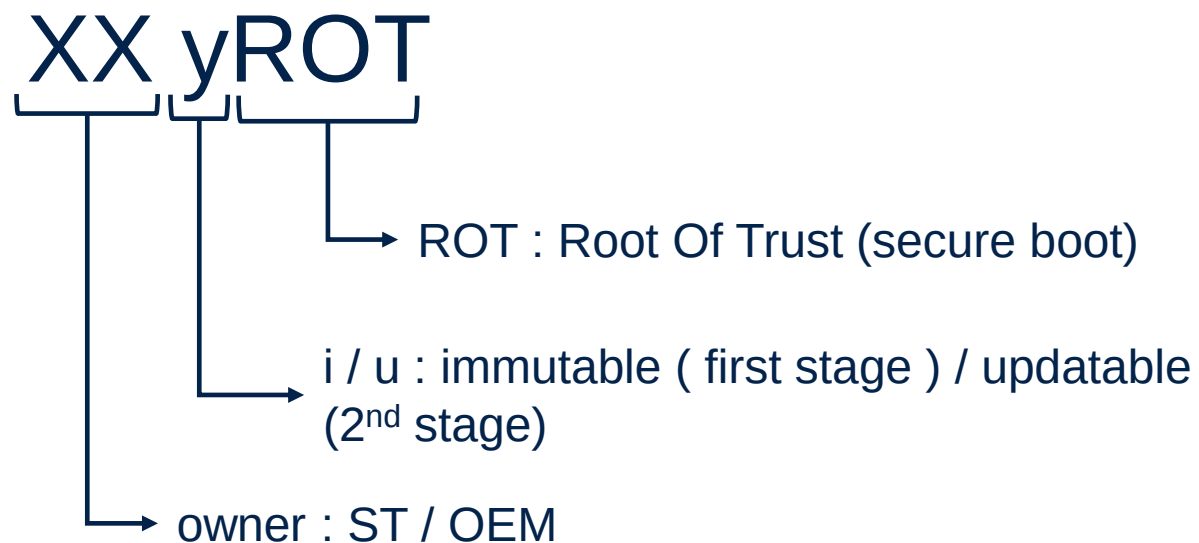
Regression

All the user flash code is erased  
OBK HDPL1,2,3S, 3NS are erased  
EPOCH\_S and EPOCH\_NS are incremented  
Similar to RDP1 to RDP0 regression

# Security bootpath



# Naming convention



- STiROT
- STuROT
- OEMiROT
- OEMuROT



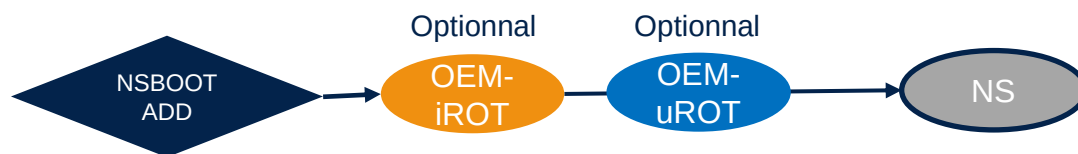
# STM32H5 Boot Path

- The STM32H5 security boot path is configured thanks to option bytes
  - TZEN : TrustZone enabled
    - Possible value : Enabled 0xB4 / Disabled 0xC3 (Values more difficult to attack !)
  - BOOT\_UBE : Unique Boot Entry selection
    - Possible value :
      - STiROT : 0xC3
      - User Flash : 0xB4 (also called OEMiROT)
    - NSBOOTADD/SECBOOTADD

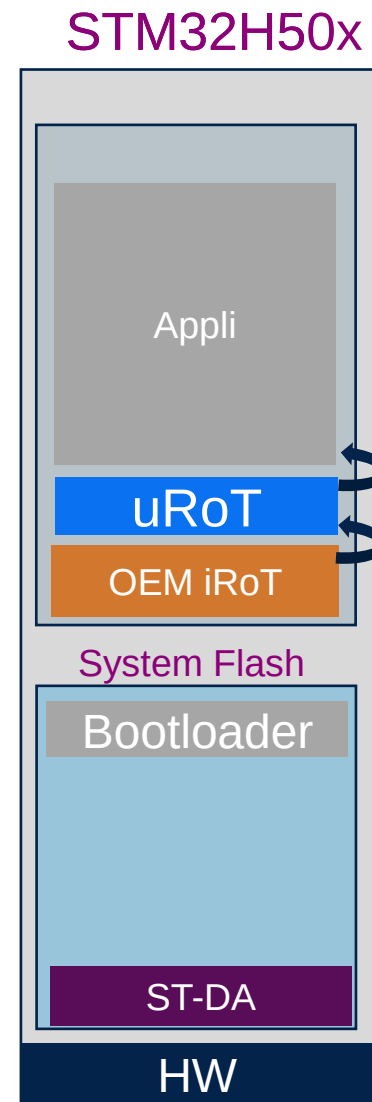


# STM32H5 Security Boot Path

- STM32H50x : no TrustZone and unique boot entry to NSBOOTADD



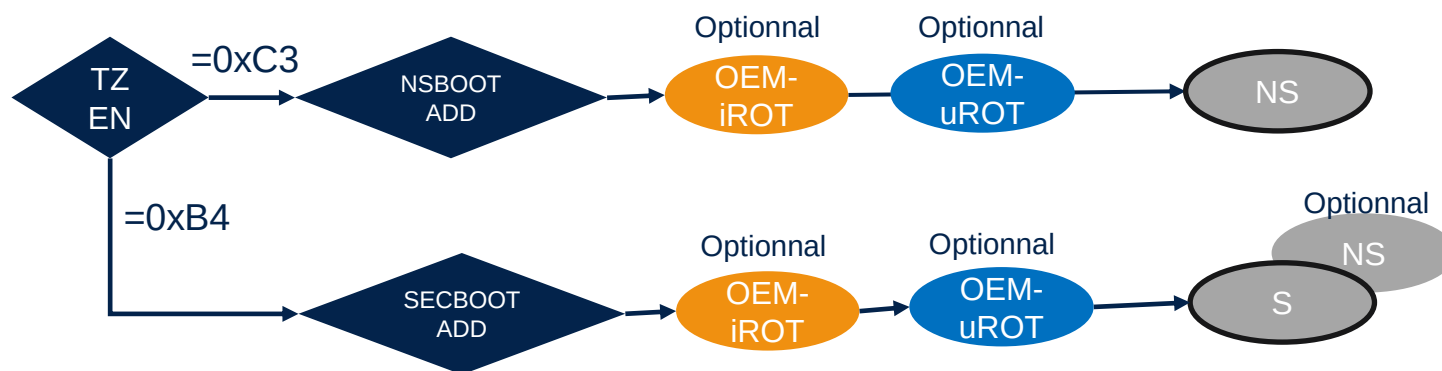
Code example available and supported by STM32CubeMx



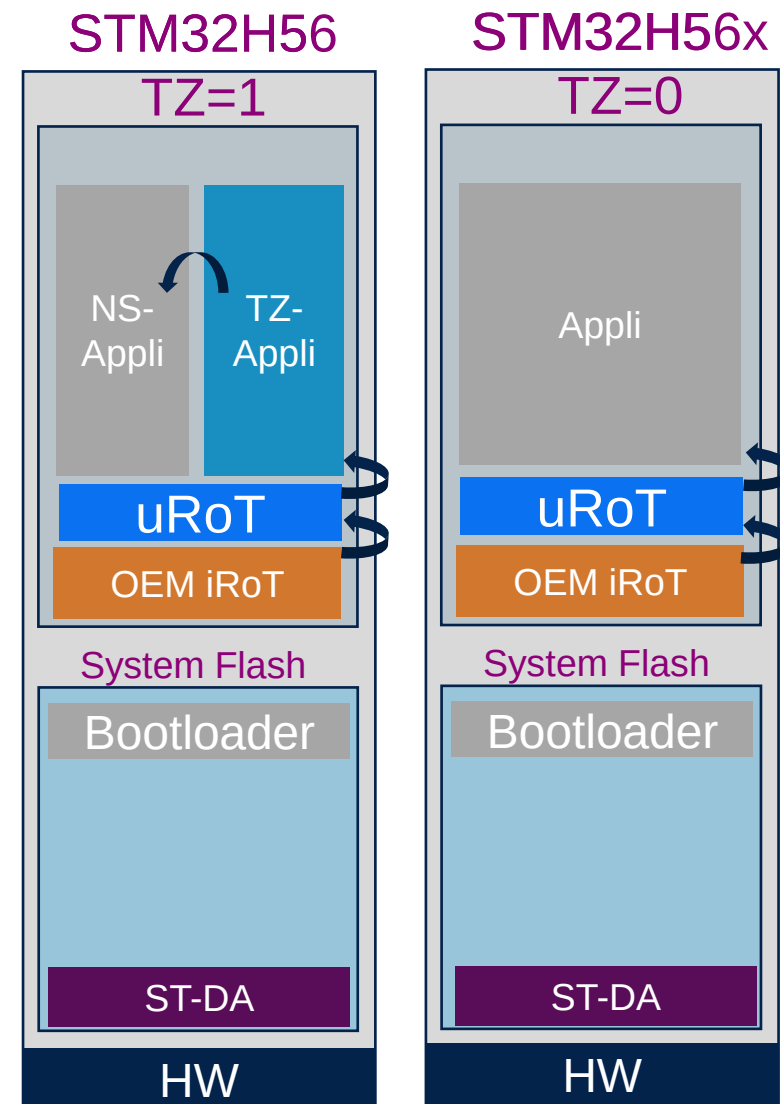


# STM32H5 Security Boot Path

- STM32H56x : TrustZone, no crypto, no STiROT



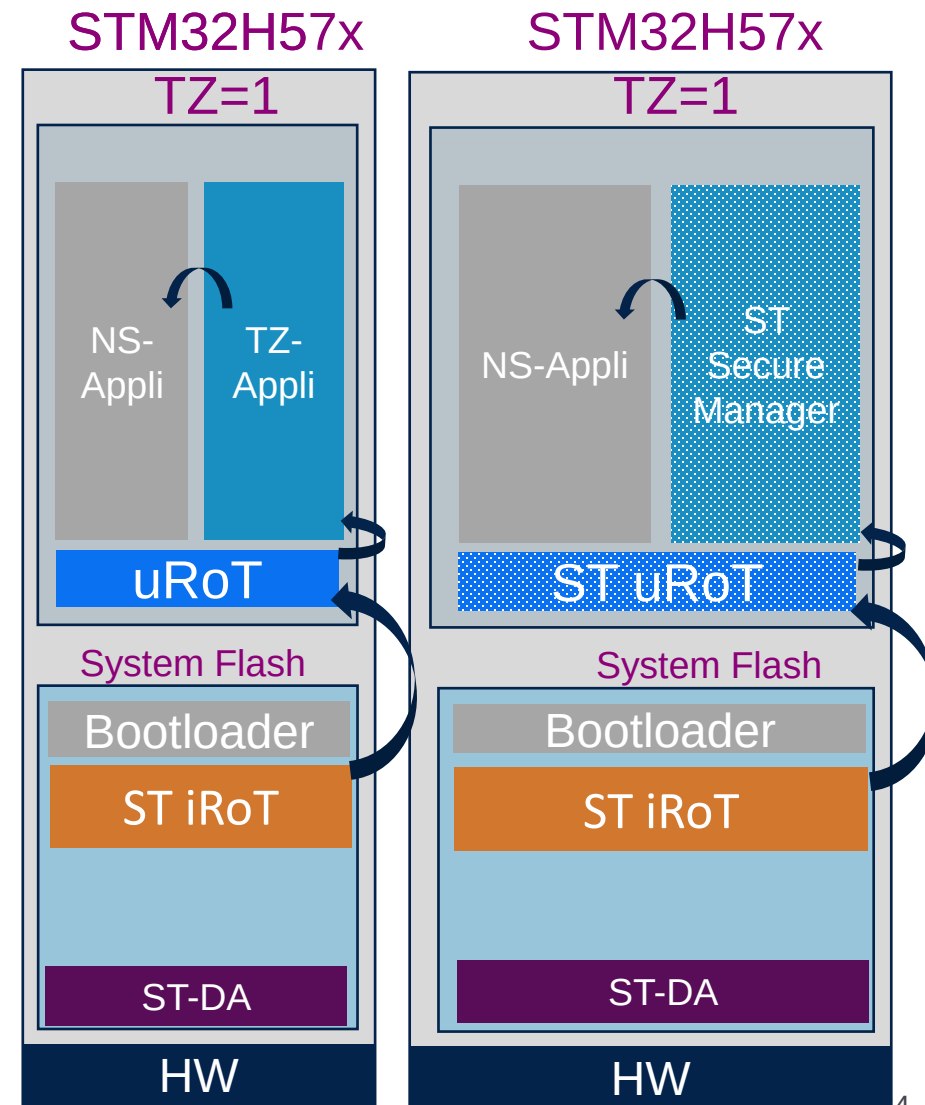
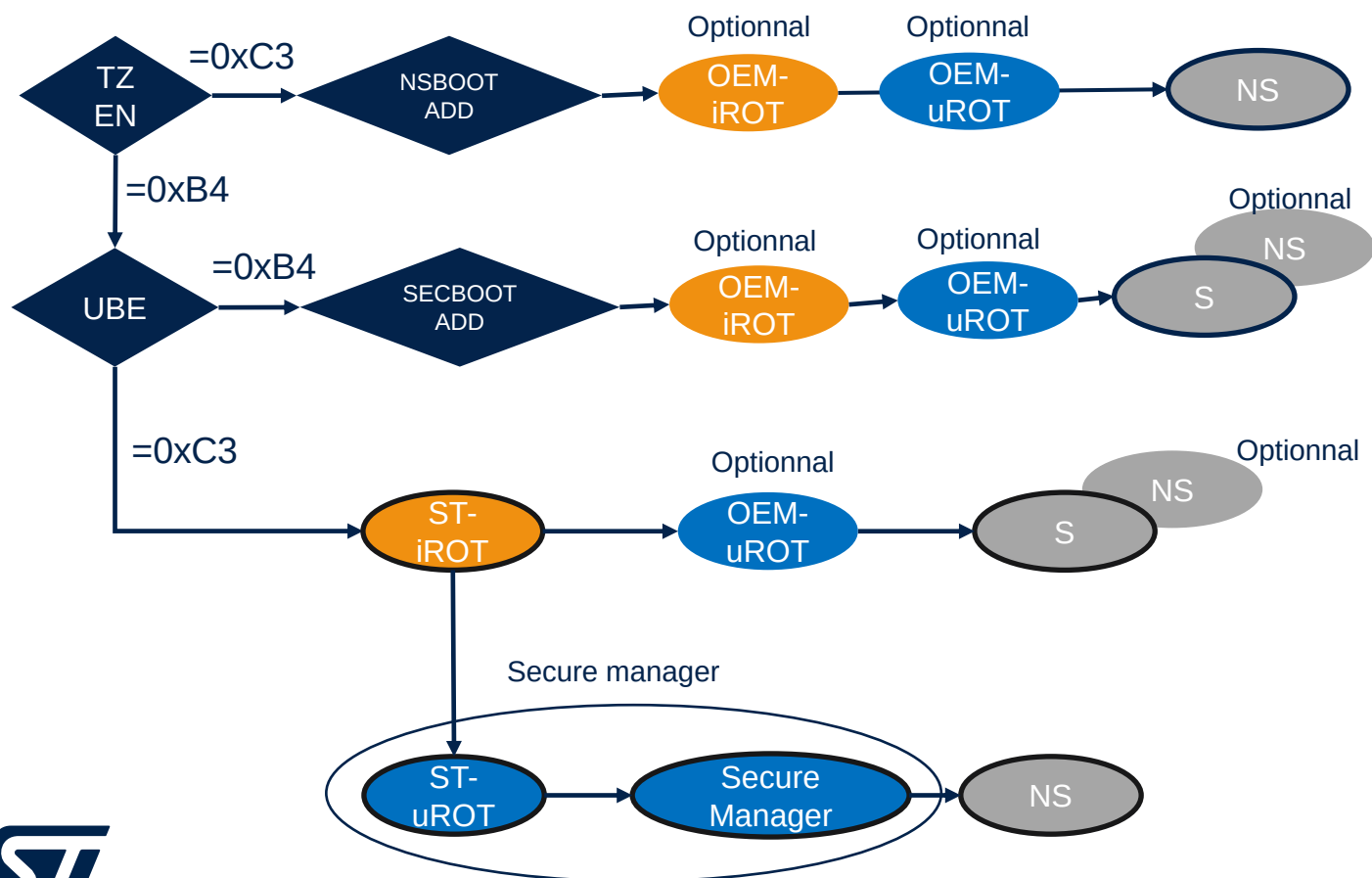
Code example available and supported by STM32CubeMx





# STM32H5 Security Boot Path

- STM32H57x : TrustZone / Crypto / STiRoT

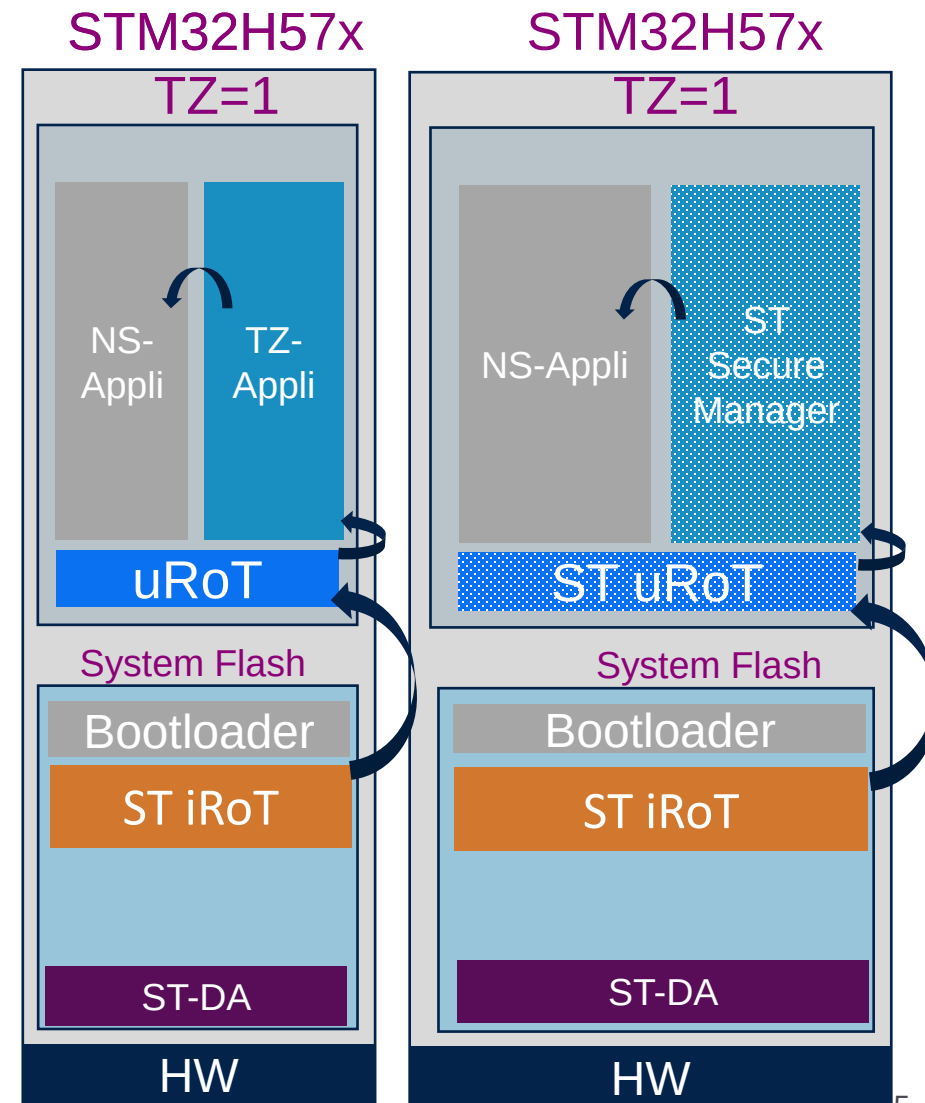
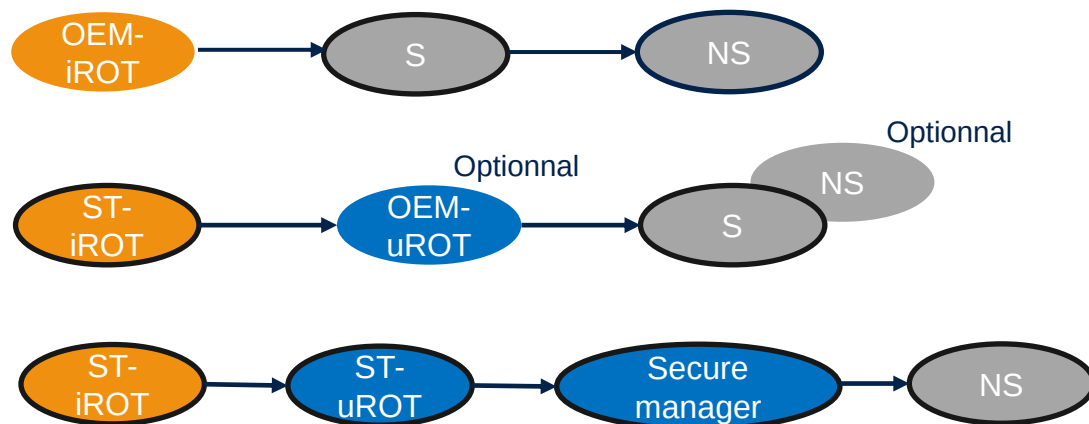




# STM32H5 Security Boot Path

- STM32H57x : TrustZone / Crypto / STiRoT

Code example available and supported by STM32CubeMx





# Security services

|                              | STM32H50x | STM32H56x                   | STM32H57x                                                                                                                                |
|------------------------------|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| RSS-Lib                      | HDPL      | HDPL<br>OBKeys provisioning | HDPL<br>OBKeys provisioning                                                                                                              |
| ST DebugAuthent              | Password  | Password/Certificate        | Password/Certificate                                                                                                                     |
| ST iRoT                      |           |                             | <br>psacertified™<br>level three<br>TZ=1<br>SESIP™3   |
| ST uRoT    ST Secure Manager |           |                             | <br>psacertified™<br>level three<br>TZ=1<br>SESIP™3 |

# STiROT



- ST immutable Root Of Trust
- Secure boot code with secure firmware update capability implemented in system flash
- Code based on MCUBoot with added countermeasures against board level attacks
- Passed SESIP Level 3 and PSA Level 3 certifications: insurance about robustness against board level attack



# How STiROT works ?

- Activated through BOOT\_UBE option byte
- Requires TrustZone enabled
- Need 2 specific provisioned data in OBKeys HDPL1
  - Configuration data (memory mapping / functionalities) and Authentication/encryption keys
  - Zero initialized data used for managing versions: dynamic



# Details of STiROT

- **Secureboot:** STiROT authenticates application in flash
  - ECC key for authentication is part of STiROT configuration stored in OBK
  - Either full secure or secure/non secure applications supported
- **Firmware update:** secure/non secure
  - **Dual slot setup** : Active slot and download slot
    - => Application can use almost half of the flash size, but not more
  - **FW Loader** : 2 possibilities (not exclusive) for downloading new code/data
    - Jump to embedded system bootloader
    - Implement loader in application code
  - One image for secure and non secure application only.



# STiRot in STM32 Ecosystem

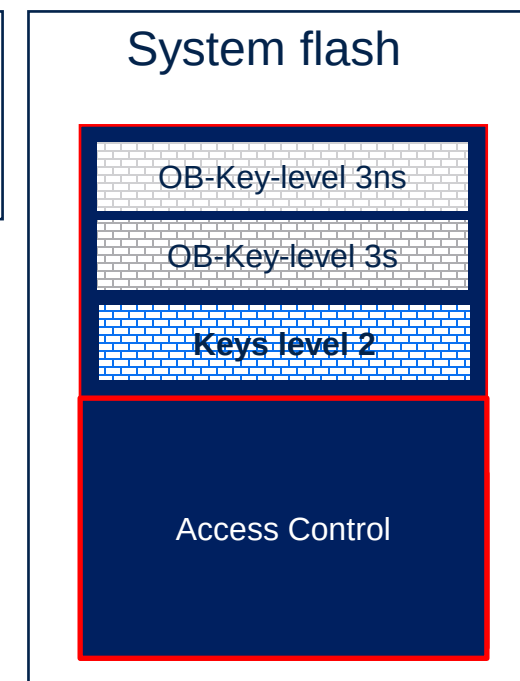
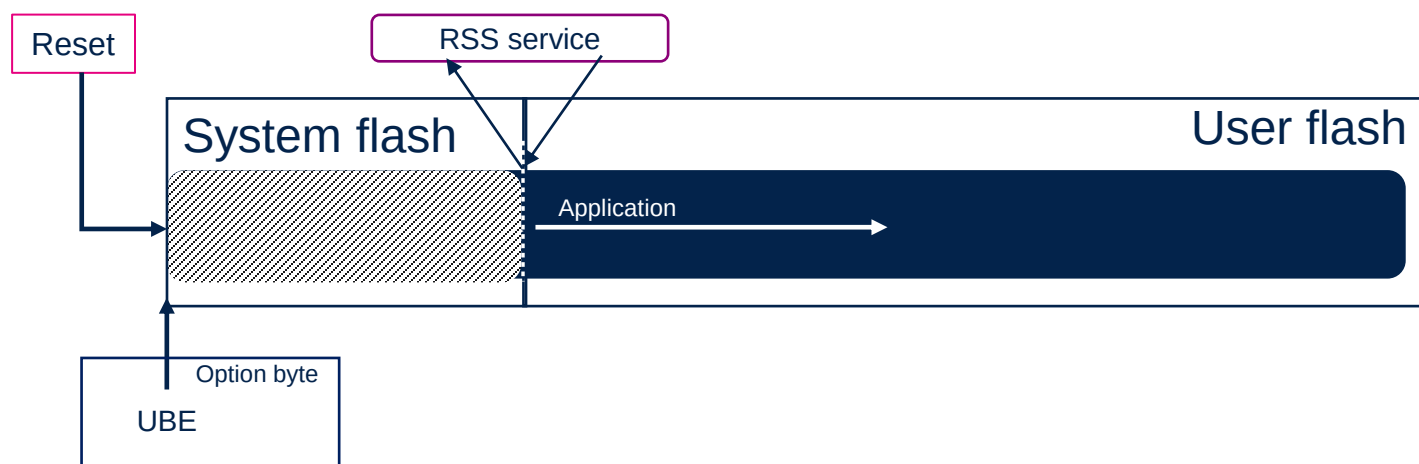
- STM32CubeH5 provides scripts and code example with STiROT.
- STM32CubeMX supports STiROT Boot Path with secure application only or with secure/non secure
- Let's see an example of CubeMX with secure only application !

# STiROT Hands-on / Demo



# STiROT boot

HDP Level = 0

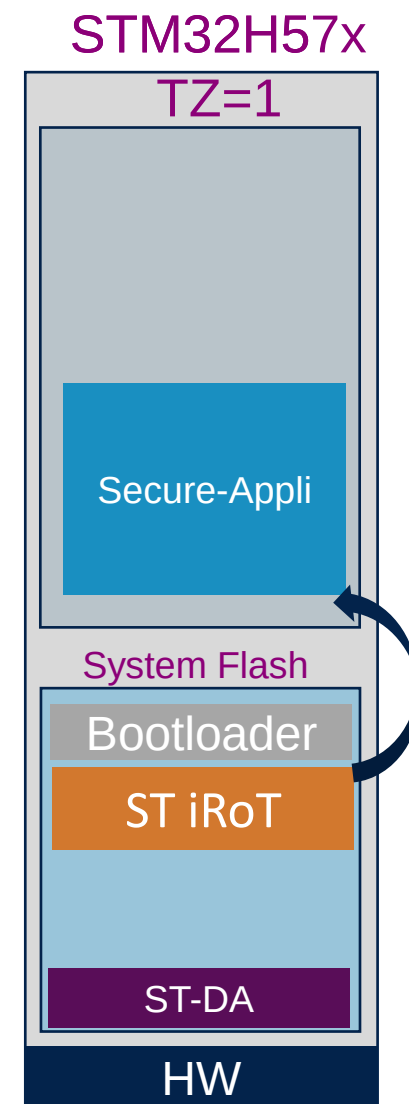


- If no application to install, check the authenticity and the integrity of the user application and if ok, launch it
- Application will be executed in HDP level 2 secure



# Hands-on purpose

- Create a secure led blinking application launched by STiROT
- When we press the button, the application will jump in the embedded bootloader.
- The embedded bootloader allows to download a new version of the application.
- On the next reset the STiROT will install this new version.





# STiROT Hands-on

- Step 0 : create an application with its associated bootpath
- Step 1 : configure STiROT
- Step 2 : develop your application
- Step 3 : provision STiROT config / DA access / Application
- Step 4 : secure update
- Step 5 : debug
- Step 6 : regression



# STEP 0 : create an application

MX STM32CubeMX Untitled

STM32CubeMX File Window Help

Home

### Existing Projects

#### Recent Opened Projects

|                                          |    |
|------------------------------------------|----|
| Test_led.ioc                             | MX |
| Last modified date : 15/12/2022 16:45:24 |    |
| BootPatch_Hands-on_with_T...             | MX |
| Last modified date : 15/12/2022 14:24:09 |    |
| Test.ioc                                 | MX |
| Last modified date : 15/12/2022 8:45:41  |    |
| GPIO_IOToggle_TrustZone.ioc              | MX |
| Last modified date : 05/12/2022 10:56:00 |    |
| STM32F769I_DISCO.ioc                     | MX |
| Last modified date : 29/11/2022 16:43:45 |    |

#### Other Projects

### New Project

I need to :

1 Start My project from MCU

ACCESS TO MCU SELECTOR

Start My project from ST Board

ACCESS TO BOARD SELECTOR

Start My project from Example

ACCESS TO EXAMPLE SELECTOR

### Manage software installations

Check for STM32CubeMX and...

CHECK FOR UPDATES

Install or remove embedded so...

INSTALL / REMOVE

MX now on v5.5

Enrich your STM32G4/F7/L5/G0/WL projects with Azure RTOS software expansion

Pinout & Configuration > Software Packs

About STM32 External Tools



# STEP 0 : create an application

**MX** New Project from a MCU/MPU

MCU/MPU Selector | Board Selector | Example Selector | Cross Selector

MCU/MPU Filters

1 **Commercial Part Number** H573II

2 **STM32H573II**

3 **Start Project**

**PRODUCT INFO**

- Segment
- Series
- Line
- Marketing Status
- Price
- Package
- Core
- Coprocessor

**MEMORY**

- Flash = 2048 (kBytes)
- EEPROM = 0 (Bytes)
- DAM Total = 640 (kBytes)

**STM32H5 Series**

**STM32H573IIK3Q** High-performance, Arm Cortex-M33 with TrustZone, MCU with 2-Mbyte Flash, 640-Kbyte RAM, 250 MHz CPU

Unit Price for 10kU (US\$) : NA

Board: [STM32H573I-DK](#)

UFBGA 176+25 10x10x0.6 P 0.65 mm

High-performance, Arm Cortex-M33 with TrustZone, MCU with 2-Mbyte Flash, 640-Kbyte RAM, 250 MHz CPU

**Features**

- N/A

MCU List: 4 items

| Commercial | Part No.       | Reference      | Marketi    | Unit Price | Board         | Package    | Flash     | RAM        |
|------------|----------------|----------------|------------|------------|---------------|------------|-----------|------------|
| ☆          | STM32H573IIK3Q | STM32H573IIKxQ | Coming ... | NA         | STM32H573I-DK | UFBGA ...  | 2048 k... | 640 kBytes |
| ☆          | STM32H573IIR6  | STM32H573IIRx  | Coming ... | NA         |               | UFBGA ...  | 2048 k... | 640 kBytes |
| ☆          | STM32H573IIT3Q | STM32H573IITxQ | Coming ... | NA         |               | LQFP 17... | 2048 k... | 640 kBytes |
| ☆          | STM32H573IIT6  | STM32H573IITx  | Coming ... | NA         |               | LQFP 17... | 2048 k... | 640 kBytes |



# STEP 0 : TrustZone activated

TrustZone feature available



Do you want to create a new project :

☐ without TrustZone activated ?

4

☒ with TrustZone activated ?

5

OK



# STEP 0 : create an application

The screenshot shows the STM32CubeMX Project Manager window. The 'Project Manager' tab is selected. The 'Project Name' field is 'SecureAppli' (annotated with a pink circle 2). The 'Project Location' field is 'C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT' (annotated with a pink circle 3). The 'Project Structure' section shows 'Secure Project' and 'Non Secure Project' both checked. The 'Application Structure' is set to 'Advanced'. The 'Toolchain Folder Location' is 'C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT\SecureAppli\'. The 'Toolchain / IDE' is 'EWARM'. The 'Min Version' is 'V9'. The 'Do not generate the main()' checkbox is unchecked. The 'Generate Under Root' checkbox is unchecked. A pink box labeled '1' highlights the 'Project Manager' tab. A pink box labeled '2' highlights the 'Project Name' field. A pink box labeled '3' highlights the 'Project Location' field. A pink arrow points from the 'SecureAppli' label to the 'Project Name' field. A pink arrow points from the 'C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT' label to the 'Project Location' field.

Pinout & Configuration

Clock Configuration

Project Manager

Tools

Project

Code Generator

Advanced Settings

Project Settings

Project Name

Project Location

Project Structure

Application Structure

Toolchain Folder Location

Toolchain / IDE

SecureAppli

C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT



# STEP 0 : Only secure application

Pinout & Configuration    Clock Configuration    **Project Manager**    Tools

Project

Code Generator

Advanced Settings

Boot Path

and Debug Authentication

Project Settings

Project Name: SecureAppli

Project Location: C:\Training\S5\_Security\_WS\Hands-on\Test\_STiROT

Project Structure: ☒ Secure Project ☐ Non Secure Project

Application Structure: ☐ Do not generate

Toolchain Folder Location: ☐ Generate Un

Toolchain / IDE: ☐ Generate Un

Linker Settings

Minimum Heap Size: ☐ Generate Un

Minimum Stack Size: ☐ Generate Un

Thread-safe Settings

CortexM33S

**Warning: Project Structure**

**You have changed your project Structure.**

- Only Secure Code will be generated and Non Secure Code won't be removed.
- If some Ips are non secured they will be ignored. Please check from the System View Tab.

OK



# STEP 0 : Select IDE

| Pinout & Configuration | Clock Configuration                                                                                               | Project Manager | Tools |
|------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------|-------|
| Project                | Project Settings                                                                                                  |                 |       |
| Code Generator         | Project Name: SecureAppli                                                                                         |                 |       |
| Advanced Settings      | Project Location: C:\Training\STM32H5_Security_WS\Hands-on\Test_STiROT <span>Browse</span>                        |                 |       |
| Boot Path              | Project Structure: <input checked="" type="checkbox"/> Secure Project <input type="checkbox"/> Non Secure Project |                 |       |
|                        | Application Structure: Advanced <span>Do not generate the main()</span>                                           |                 |       |
|                        | Toolchain Folder Location: C:\Training\STM32H5_Workshop\Hands-on\Test_ST_iROT\SecureAppli\                        |                 |       |
|                        | Toolchain / IDE: <b>STM32CubeIDE</b> <input checked="" type="checkbox"/> Generate Under Root                      |                 |       |

1



# STEP 0 : Save Project

1

2

File

Window

Help

Hello Frantz

10

f

YouTube

Twitter

GitHub

ST

Project Manager

GENERATE CODE

Block Configuration

Project Manager

Tools

SecureAppli

C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT

Browse

☒ Secure Project ☐ Non Secure Project

Advanced

☐ Do not generate the main()

C:\Training\STM32H5\_Workshop\Hands-on\Test\_ST\_iROT\SecureAppli\

STM32CubeIDE

☒ Generate Under Root

Linker Settings

|                    | M33S  | M33NS |
|--------------------|-------|-------|
| Minimum Heap Size  | 0x200 | 0x200 |
| Minimum Stack Size | 0x400 | 0x400 |



# STEP 0 : Setup the bootpath

Pinout & Configuration

Project

Code Generator

Advanced Settings

Boot Path  
and Debug Authentication

Clock Configuration

Debug Authentication  
Configure

Boot Path Selection  
2 Select No boot path selected (Nothing to configure) 

Application (Default Boot Path)

Project Manager



# STEP 0 : Select STiROT

MX Bootpath



Define your bootpath

TZEN=1 'Only Secure' project

Boot Step#1

☐ Secure Application

☐ OEM immutable Root of Trust (OEM-iRoT)

☐ ST immutable Root of Trust (ST-iRoT)

Graphical Boot Representation

1

BACK

NEXT

FINISH

CANCEL

MX Boot Path



Define your boot Path

TZEN=1 'Only Secure' project

Boot Step#1

☐ Secure Application

☐ OEM immutable Root of Trust (OEM-iRoT)

☒ ST immutable Root of Trust (ST-iRoT)

Graphical Boot Representation

| Non Secure | Secure  |
|------------|---------|
|            |         |
|            |         |
|            | ST-iRoT |

STM32

2

BACK

NEXT

FINISH

CANCEL



# STEP 0 : Select STiROT and Secure Application

MX Boot Path

ST Define your boot Path

TZEN=1 'Only Secure' project

**1** Boot Step#1

☐ Secure Application

☐ OEM immutable Root of Trust (OEM-iRoT)

☒ ST immutable Root of Trust (ST-iRoT)

Graphical Boot Representation

| Non Secure | Secure  |
|------------|---------|
|            |         |
|            | ST-iRoT |
| STM32      |         |

BACK NEXT FINISH CANCEL

MX Boot Path

ST Define your boot Path

TZEN=1 'Only Secure' project

Boot Step#2

☒ Secure Application

☐ updatable Root of Trust (uRoT)

Graphical Boot Representation

| Non Secure | Secure             |
|------------|--------------------|
|            | Secure Application |
|            | ST-iRoT            |
| STM32      |                    |

**2**

BACK NEXT **FINISH** CANCEL

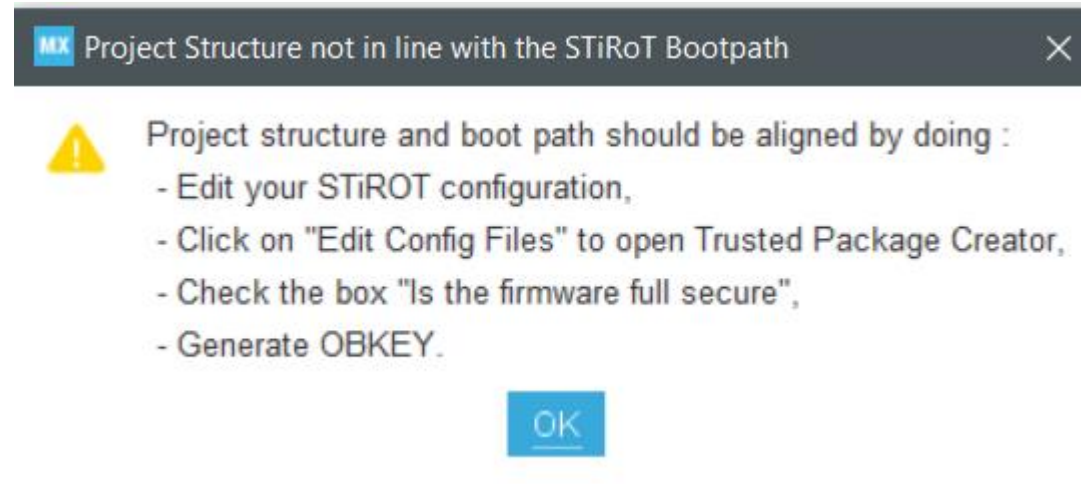


# STEP 0 : Information

| Pinout & Configuration | Clock Configuration                                                                                                                                                                                                                                                                                                                                                                                               | Project Manager                             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Project                | Debug Authentication<br><input type="button" value="Configure"/> <input checked="" type="checkbox"/> Generate DA Folder                                                                                                                                                                                                                                                                                           | <div>Only with CubeMX 6.10.0 or upper</div> |
| Code Generator         | Boot Path Selection<br><input type="button" value="Select"/> ST_IROT                                                                                                                                                                                                                                                                                                                                              |                                             |
| Advanced Settings      | <input checked="" type="checkbox"/> Use Default Reference Location<br>Reference Location <input type="text" value="C:\STM32Cube_FW_H5_V1.1.1\Projects\STM32H573I-DK\ROT_Provisioning\STiROT\"/> <input type="button" value="Browse"/><br>Config Files Location <input type="text" value="C:\Training\STM32H5_Security_WS\Hands-on\pip\ROT_Provisioning\STiROT"/> <input type="button" value="Edit Config Files"/> |                                             |



# STEP 0 : Information





# STiROT Hands-on

- ~~Step 0 : create an application with associated its bootpath~~
- Step 1 : configure STiROT
- Step 2 : develop your application
- Step 3 : provision STiROT config / DA access / Application
- Step 4 : secure update
- Step 5 : debug
- Step 6 : regression

## Step 2 : configure STiROT



# STEP 1 : Application will signed and encrypted

Debug Authentication

Configure

Boot Path Selection

Select ST\_IROT

☒ Use Default Reference Location

Reference Location

C:\Users\lefrere\STM32Cube\Repository\STM32Cube\_FW\_H5\_V1.1.1\Projects\STM32H573I-DK\IROT\_Provisioning\STIROT

Browse

Config Files Location

C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STIROT\SecurityApp\IROT\_Provisioning\STIROT

Edit Config Files

Linker

☒ Enable Linker File(s) Update  
☐ Reinitialize Linker File(s)

Signature

☒ Sign Binary(ies)

Mapping

M33S

Start address0xC000400End address0xC01F3BF

Boot Path Image

| Non Secure | Secure             |
|------------|--------------------|
|            | Secure Application |
|            | ST-IrOT            |

STM32



# STEP 1 : STiROT configuration

1

 STM32 Trusted Package Creator

File

Edit

Options

Help

Doc



H5-Image Gen 1

OBkey

XML file

g:/STM32H5\_Workshop/Hands-on/Test\_ST\_iROT/SecureAppli/ROT\_Provisioning/STiROT/Config/STiROT\_Config.xml

Open



Number of images managed

0x01 | firmware image only



Is the firmware full secure

☒

Firmware execution area offset

0x00000000

Firmware download area offset

0x001E0000

Firmware area size

0x00020000

Data download area offset

Size of the secure area inside the firmware execution area

Product state minimal allowed

0x0000ED00 | Open



Encryption key

.\Keys\STiROT\_Encryption.pem

Open

ecdsa-p256

Regenerate

Authentication key

.\Keys\STiROT\_Authentication.pem

Open

ecdsa-p256

Regenerate

STiROT configuration

Information

OBkey Destination Address

0x0FFD0200

Encryption Option

enable

Slot Information

| Index | Name                    | Address    |
|-------|-------------------------|------------|
| 0     | Firmware execution area | 0xc0000000 |

Output File

./../Binary/STiROT\_Config.obk

Select Path

The purpose of this tool is to generate provisioning data also called obk files.  
The file can be generated only with XML input.  
Possible XML configuration for OBKey are:

- 1.DA (Debug Authentication) configuration XML
- 2.STiROT configuration XML
- 3.OEMiROT configuration XML
- 4.OEMuROT configuration XML
- 5.SMAK configuration XML

-Examples of XML can be found in CubeFW package in projects\<board>\ROT\_Provisioning



Encryption option is enabled! Make sure that you use crypto part (STM32H573)!

Generate OBkey



# STEP 1 : STiROT configuration

- Implemented as a xml file
- Located in ROT\_Provisioning\STiROT\Config\STiRoT\_Config.xml
- Format read by TrustedPackageCreator
- Provides easy to use GUI for modification
- GenerateOBK button will create STiRot\_Config.obk in ROT\_Provisioning\STiROT\Binary
- For this workshop, we stay with default configuration. No need to launch the obk generation again.



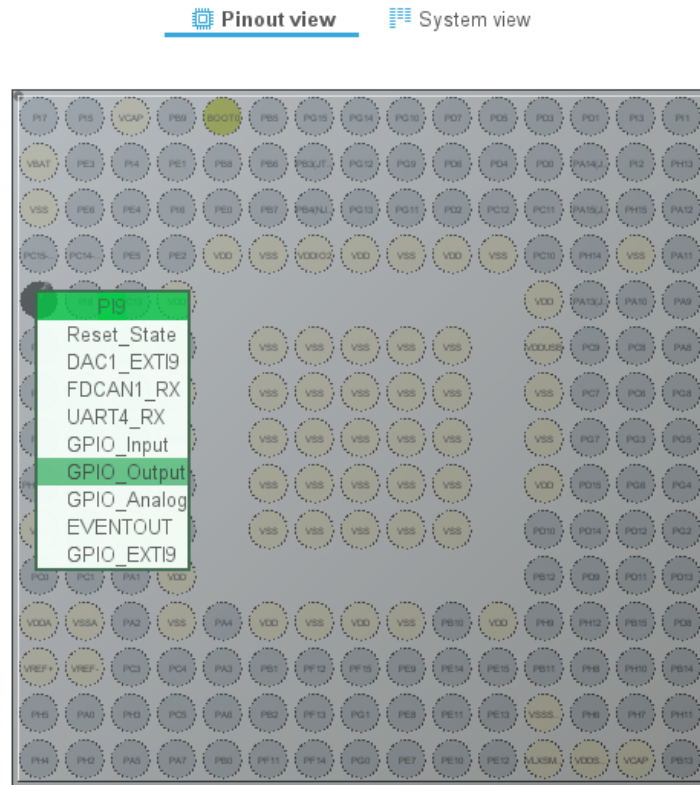
# STiROT Hands-on

- ~~Step 0 : create an application with associated its bootpath~~
- ~~Step 1 : configure STiROT~~
- Step 2 : develop your application
- Step 3 : provision STiROT config / DA access / Application
- Step 4 : secure update
- Step 5 : debug
- Step 6 : regression

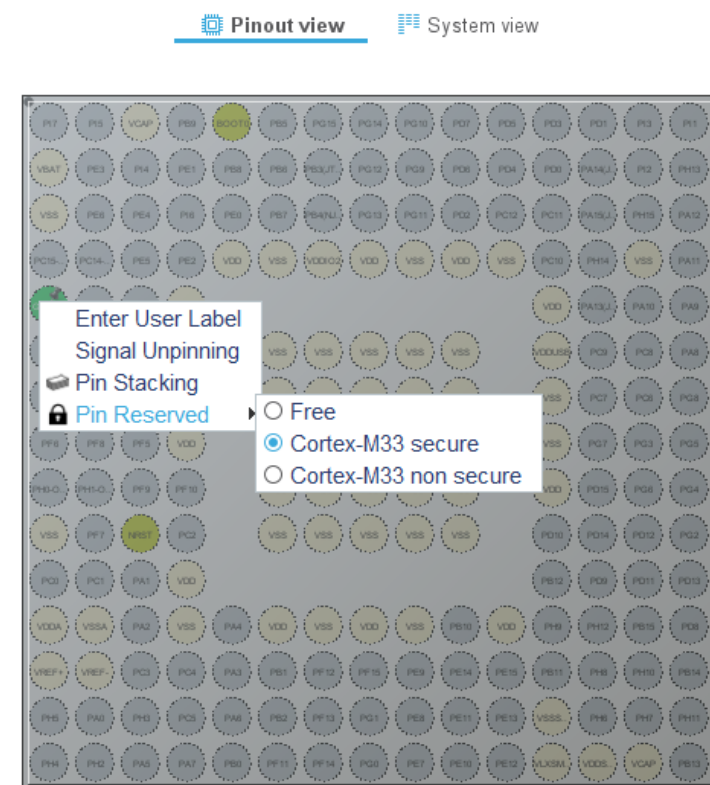
## **Step 2 : develop your application**



# STEP 2 : Configure LED pin P19 as output/secure



UFBGA176 +25 (Top view)

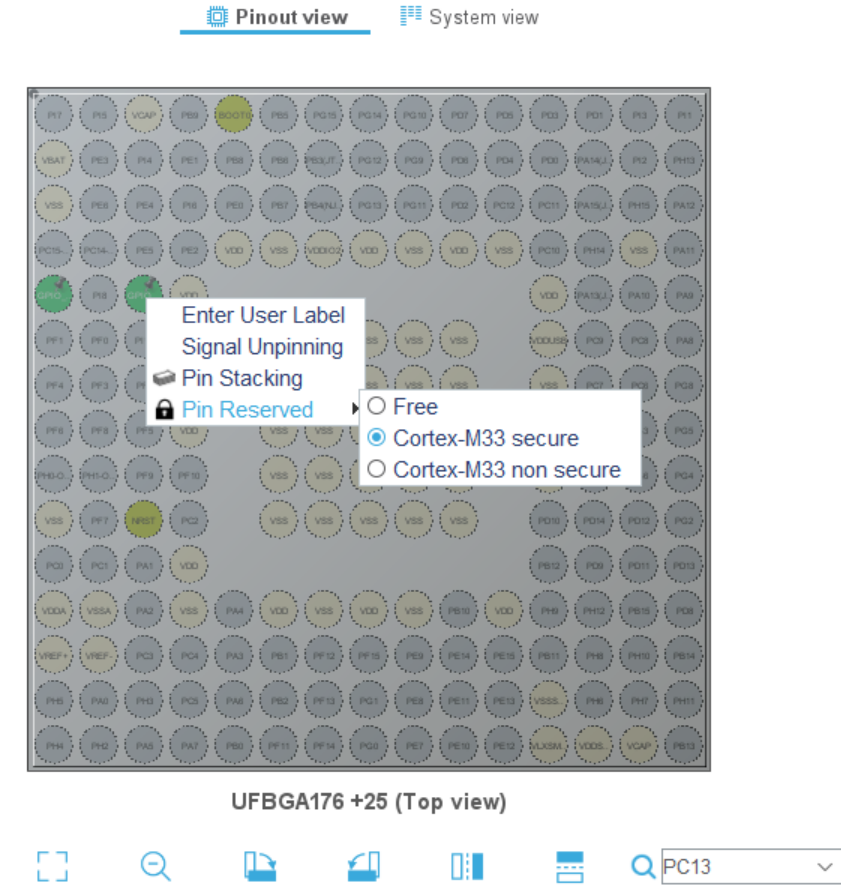
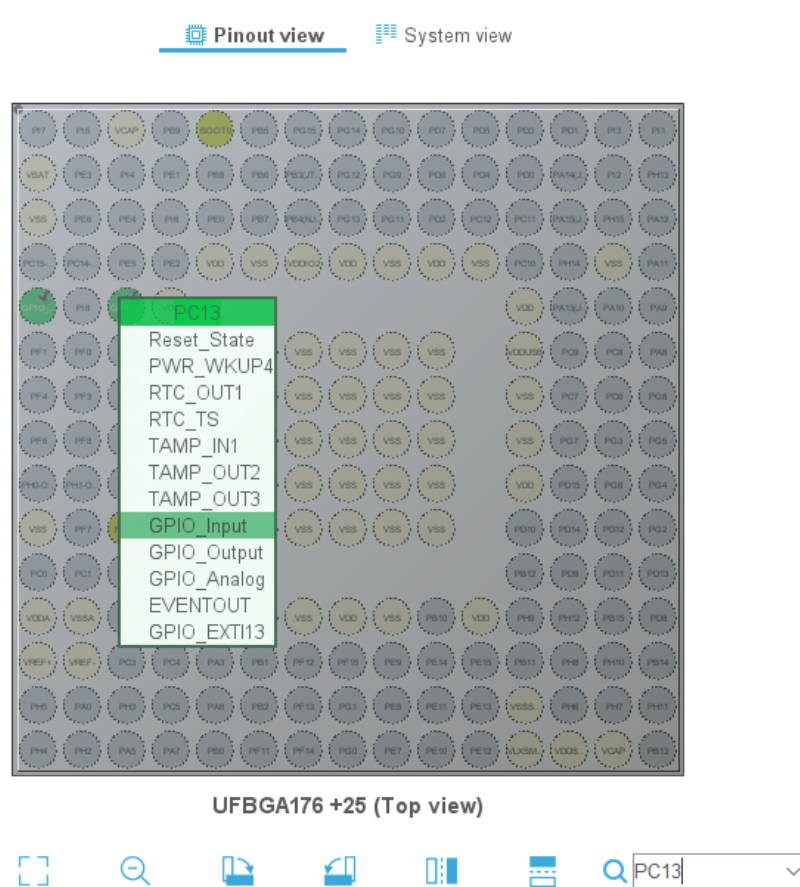


UFBGA176 +25 (Top view)





# STEP 2 : Configure USER Button pin PC13 as input/secure





# STEP 2 : Check configuration

Categories A->Z

System Core

|               | M33S                             | M33NS                            |
|---------------|----------------------------------|----------------------------------|
| CORTEX_M33    |                                  | <input checked="" type="radio"/> |
| CORTEX_M33... | <input checked="" type="radio"/> |                                  |
| DCACHE1       | <input type="radio"/>            | <input type="radio"/>            |
| GPDMA1        | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| GPDMA2        | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <b>GPIO</b>   |                                  |                                  |
| ICACHE        | <input type="radio"/>            | <input type="radio"/>            |
| IWDG          | <input type="radio"/>            | <input type="radio"/>            |

Configuration

Group By Peripherals

☒ GPIO

Search Signals  
Search (Ctrl+F)

☐ Show only Modified Pins

| Pin ... | Signal ... | Pin Context Assign | Pin Pr... | GPIO | GPIO mode        | GPI... | Ma... | Fast... | Us... | Mo...                               |
|---------|------------|--------------------|-----------|------|------------------|--------|-------|---------|-------|-------------------------------------|
| PC13    | n/a        | Cortex-M33 secure  | n/a       | n/a  | Input mode       | No ... | n/a   | n/a     |       | <input checked="" type="checkbox"/> |
| PI9     | n/a        | Cortex-M33 secure  | n/a       | Low  | Output Push Pull | No ... | Low   | n/a     |       | <input checked="" type="checkbox"/> |



# STEP 2 : Generate the code

SecureAppli.ioc - Pinout & Configuration

GENERATE CODE

Pinout view System view

Warning: Code Generation

WARNINGS:

- The STIROT selected configuration is not aligned with your selected project structure. Please go back to ProjectManager tab and check your project struture and boot path configuration.
- The instruction cache (ICACHE) must be enabled to reach the maximum performance. The ICACHE can be enabled from the Pinout tab under ICACHE.

Do you still want to generate code ?

Yes No

UFBGA176 +25 (Top view)

PC13



# STEP 2 : Generate the code

SecureAppli.ioc - Pinout & Configuration GENERATE CODE

Configuration Clock Configuration Project Manager Tools

Software Packs Pinout

Pinout view System view

Code Generation

The Code is successfully generated under :  
C:/Training/STM32H5\_Workshop/Hands-on/Test\_ST\_iROT/SecureAppli  
Project language : C

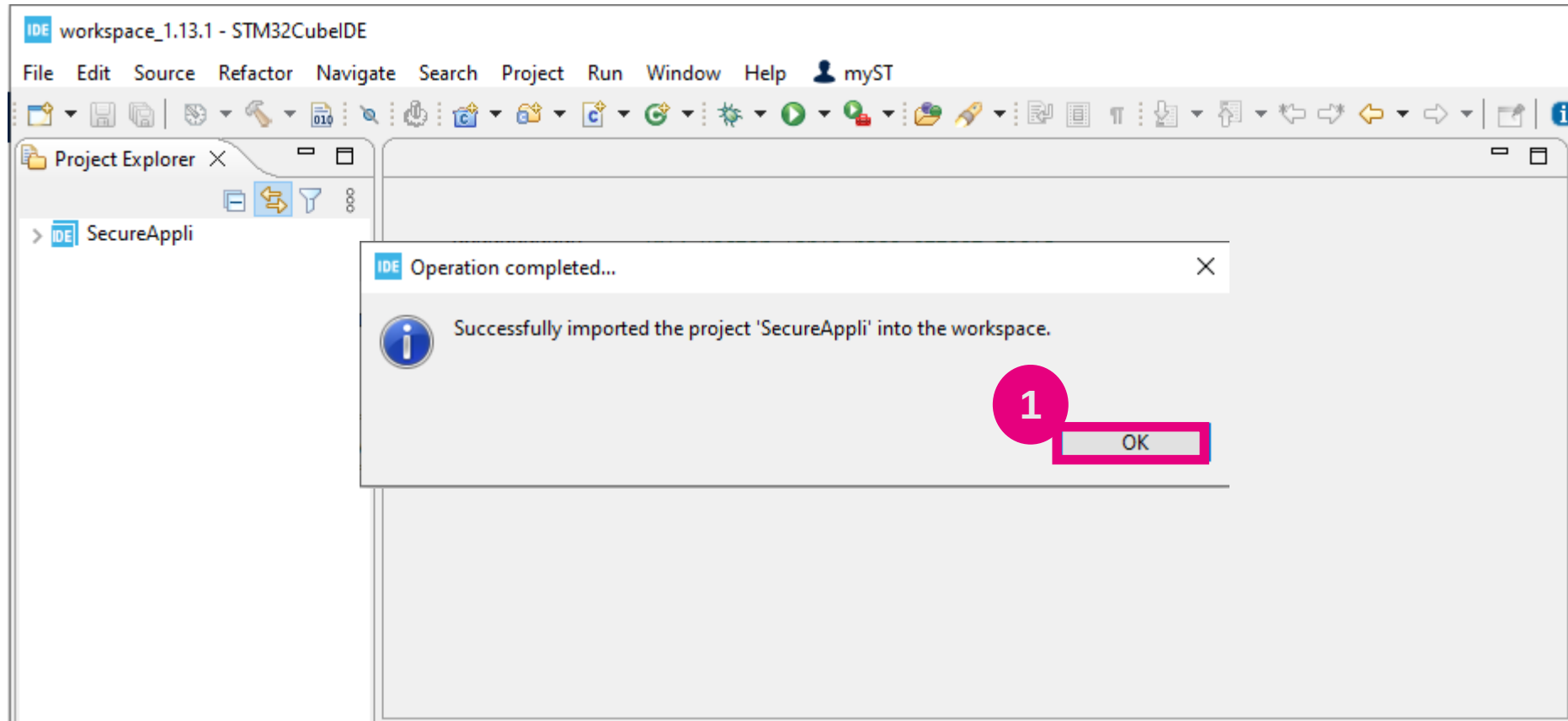
1

Open Folder Open Project Close

UFBGA176 +25 (Top view)

PC13

# STEP 2 : Generate the code



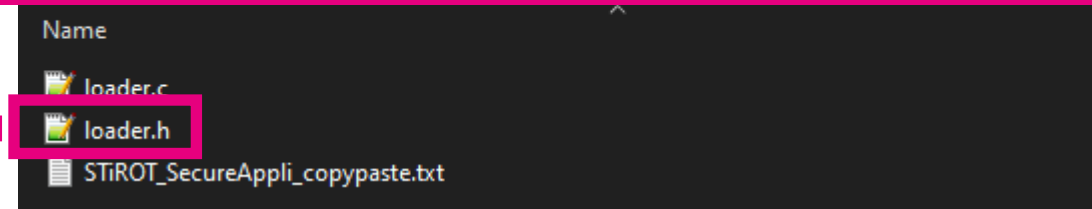
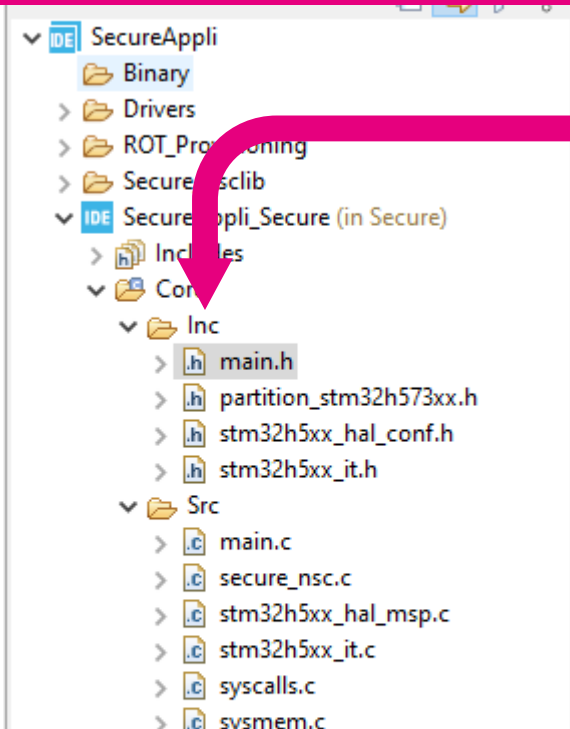


IDE

## STEP 2 : Add the code to jump in bootloader

1

Drag and drop: loader.h  
In Core\Inc folder



2

Select how files should be imported into the project:

☒ Copy files

☐ Link to files

☒ Create link locations relative to: PROJECT\_LOC

[Configure Drag and Drop Settings](#)

3

OK

Cancel

Material location :  
C:\Training\STM32H5\_Security\_WS\Hands-on-material\  
STiROT-Hands-on\



IDE

## STEP 2 : Add the code to jump in bootloader

1

Drag and drop: loader.c  
In Core\Src folder

Project Explorer X

- > Includes
- > Core
  - > Inc
    - loader.h
    - main.h
    - partition\_stm32h573xx.h
    - stm32h5xx\_hal\_conf.h
    - stm32h5xx\_it.h
  - > Src
    - main.c
    - secure\_nsc.c
    - stm32h5xx\_hal\_msp.c
    - stm32h5xx\_it.c
    - syscalls.c
    - systemem.c
    - system\_stm32h5xx\_s.c
  - > Startup
  - > Drivers
    - STM32H573IIKXQ\_FLASH.Id
    - STM32H573IIKXQ\_RAM.Id

Name

- loader.c
- loader.h
- STiROT\_SecureAppli\_copypaste.txt

IDE File Operation

2 Select how files should be imported into the project:

☒ Copy files  
☐ Link to files

☒ Create link locations relative to: PROJECT\_LOC

[Configure Drag and Drop Set](#)



3

OK

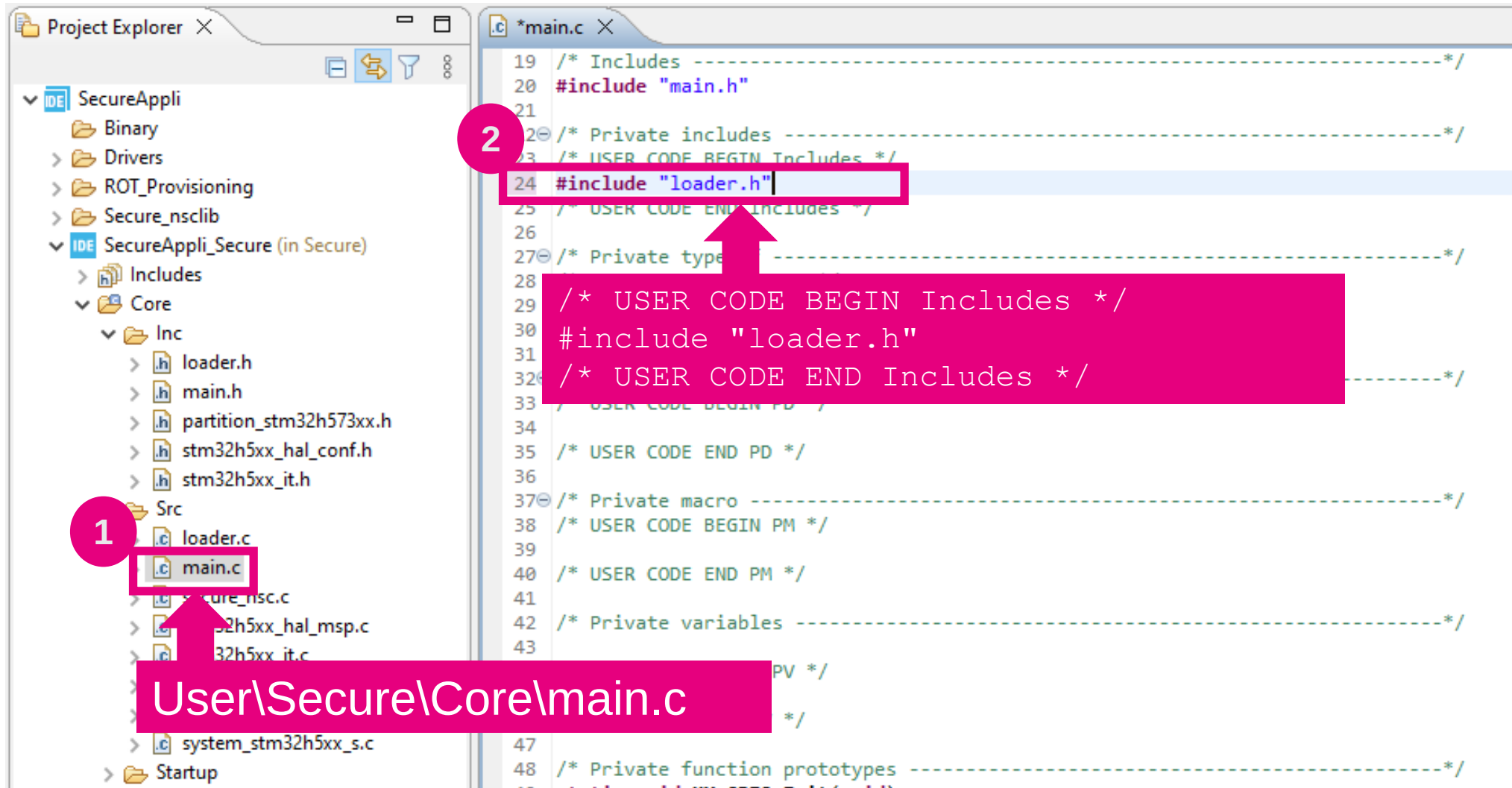
Cancel

Project Explorer X

- > IDE SecureAppli
  - > Binary
  - > Drivers
  - > ROT\_Provisioning
  - > Secure\_nsclib
  - > IDE SecureAppli\_Secure (in Secure)
    - > Includes
    - > Core
      - > Inc
        - loader.h
        - main.h
        - partition\_stm32h573xx.h
        - stm32h5xx\_hal\_conf.h
        - stm32h5xx\_it.h
      - > Src
        - loader.c
        - main.c
        - secure\_nsc.c
        - stm32h5xx\_hal\_msp.c
        - stm32h5xx\_it.c
        - syscalls.c
        - systemem.c
        - system\_stm32h5xx\_s.c
      - > Startup

Material location :  
C:\Training\STM32H5\_Security\_WS\Hands-on-material\  
STiROT-Hands-on\

# STEP 2 : Add led blinking code and jump in bootloader



Project Explorer

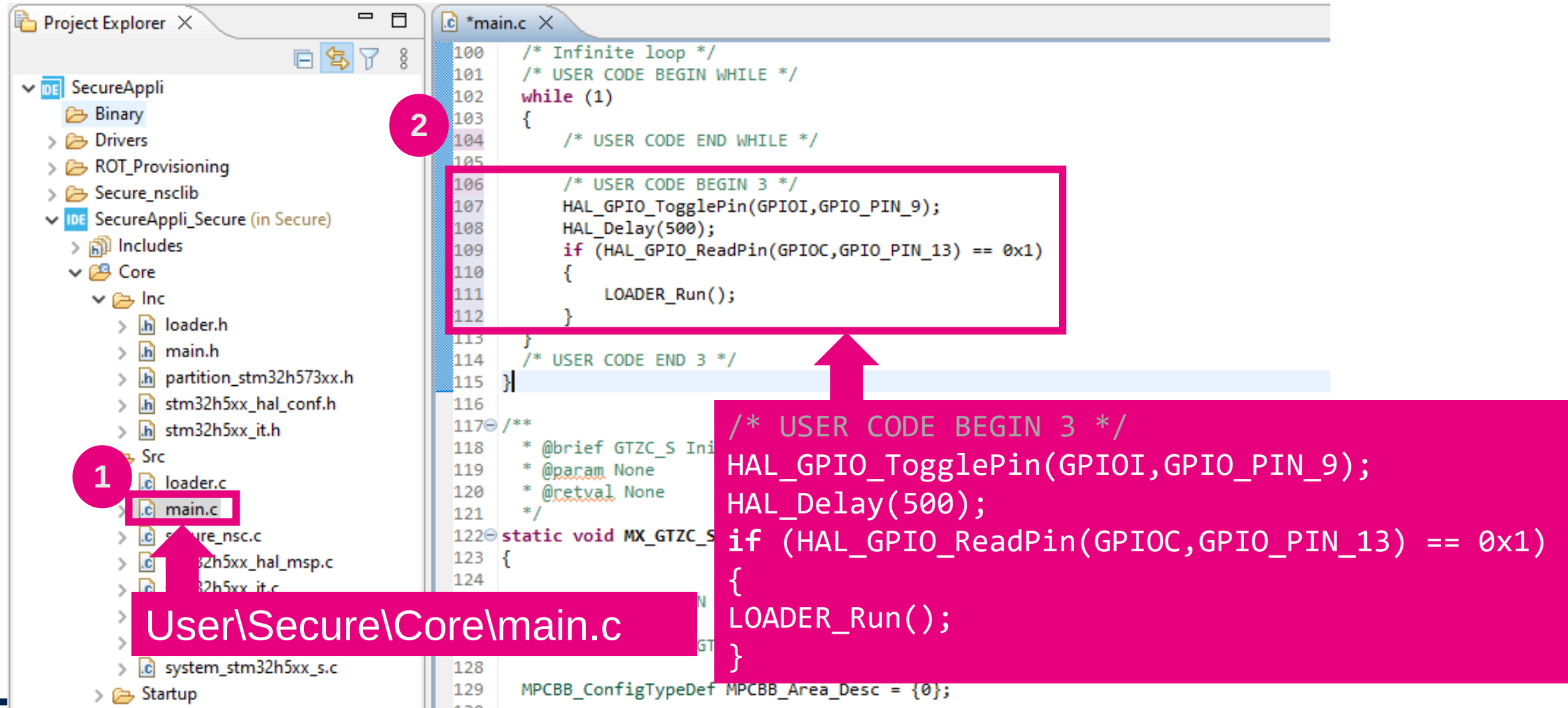
- IDE SecureAppli
  - Binary
  - Drivers
  - ROT\_Provisioning
  - Secure\_nsclib
  - IDE SecureAppli\_Secure (in Secure)
    - Includes
    - Core
      - Inc
        - loader.h
        - main.h
        - partition\_stm32h573xx.h
        - stm32h5xx\_hal\_conf.h
        - stm32h5xx\_it.h
      - Src
        - loader.c
        - main.c**
        - secure\_nsc.c
        - stm32h5xx\_hal\_msp.c
        - stm32h5xx\_it.c
      - system\_stm32h5xx\_s.c
      - Startup

User\Secure\Core\main.c

```

19 /* Includes ----- */
20 #include "main.h"
21
22 /* Private includes ----- */
23 /* USER CODE BEGIN Includes */
24 #include "loader.h"
25 /* USER CODE END Includes */
26
27 /* Private type ----- */
28
29 /* USER CODE BEGIN Includes */
30 #include "loader.h"
31 /* USER CODE END Includes */
32
33 /* USER CODE BEGIN PD */
34
35 /* USER CODE END PD */
36
37 /* Private macro ----- */
38 /* USER CODE BEGIN PM */
39
40 /* USER CODE END PM */
41
42 /* Private variables ----- */
43
44 /* Private function prototypes ----- */
45
46
47
48 /* Private function prototypes ----- */
  
```

# STEP 2 : Add led blinking code and jump in bootloader



1

2

User\Secure\Core\main.c

```

100  /* Infinite loop */
101  /* USER CODE BEGIN WHILE */
102  while (1)
103  {
104      /* USER CODE END WHILE */
105
106      /* USER CODE BEGIN 3 */
107      HAL_GPIO_TogglePin(GPIOI,GPIO_PIN_9);
108      HAL_Delay(500);
109      if (HAL_GPIO_ReadPin(GPIOC,GPIO_PIN_13) == 0x1)
110      {
111          LOADER_Run();
112      }
113  }
114  /* USER CODE END 3 */
115 }
116
117 /**
118  * @brief GTZC_S Init
119  * @param None
120  * @retval None
121  */
122 static void MX_GTZC_S_Init(void)
123 {
124
125
126
127
128
129  MPCBB_ConfigTypeDef MPCBB_Area_Desc = {0};
130

```

```

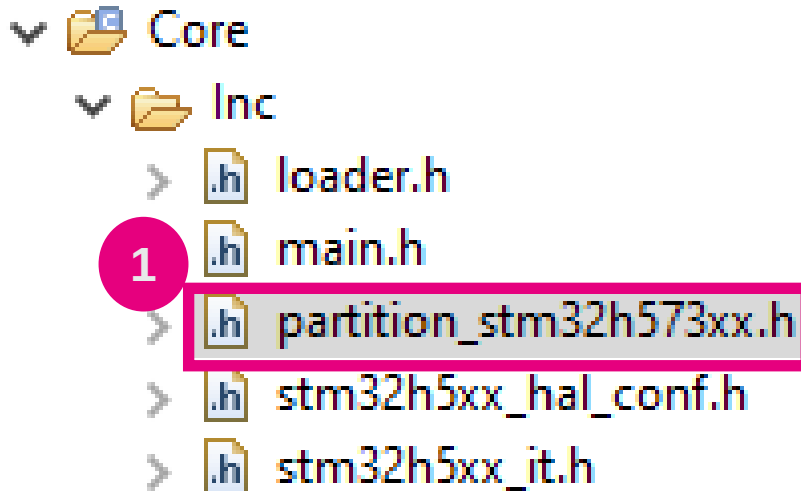
/* USER CODE BEGIN 3 */
HAL_GPIO_TogglePin(GPIOI,GPIO_PIN_9);
HAL_Delay(500);
if (HAL_GPIO_ReadPin(GPIOC,GPIO_PIN_13) == 0x1)
{
    LOADER_Run();
}

```



# STEP 0 : patch if you use CubeMX 6.10.0 or upper

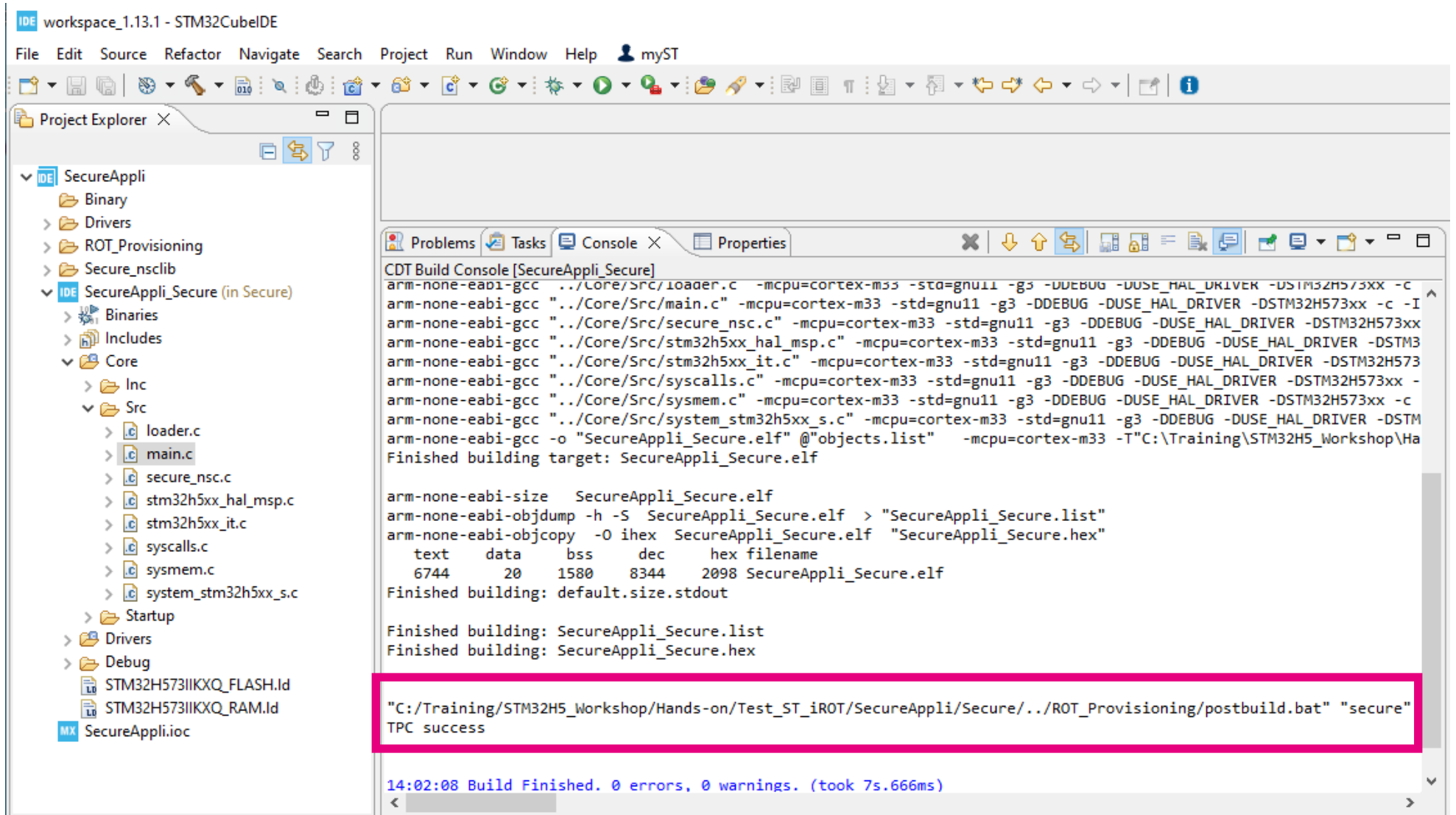
Only with CubeMX 6.10.0 or upper



```
partition_stm32h573xx.h X
205
206 /*
207 //  <e>Initialize SAU Region 5
208 //  <i>Setup SAU Region 5 memory attributes
209 */
210 #define SAU_INIT_REGION5    0
211
212 /*
213 //  <o>Start Address <0-0xFFFFFFFF>
214 */
215 #define SAU_INIT_START5    0x0BF90000    /* start address of SAU region 5 */
216
217 /*
218 //  <o>End Address <0x1F-0xFFFFFFFF>
219 */
220 #define SAU_INIT_END5      0x0BFA8FFF    /* end address of SAU region 5 */
221
222 /*
223 //  <o>Region is
224 //      <0=>Non-Secure
225 //      <1=>Secure, Non-Secure Callable
226 */
227 #define SAU_INIT_NSC5      0
```

Deactivate SAU region 5  
Line 210 : Change 1 to 0

# STEP 2 : Compile and postbuild succeeded



The screenshot shows the STM32CubeIDE interface with the Project Explorer on the left and the Console window on the right. The Project Explorer displays the project structure for 'SecureAppli', including folders like 'Binary', 'Drivers', 'ROT\_Provisioning', 'Secure\_nscib', and 'SecureAppli\_Secure (in Secure)'. The Console window shows the output of the CDT Build Console, indicating that the build for 'SecureAppli\_Secure' was successful. The output includes compilation commands for various source files, linking, and the generation of the final ELF, list, and hex files. A pink box highlights the postbuild step, which executed 'postbuild.bat' successfully. The console also shows the build finished with 0 errors and 0 warnings, taking 7s.666ms.

```

IDE workspace_1.13.1 - STM32CubeIDE
File Edit Source Refactor Navigate Search Project Run Window Help myST

Project Explorer X
  IDE SecureAppli
    Binary
    Drivers
    ROT_Provisioning
    Secure_nscib
    IDE SecureAppli_Secure (in Secure)
      Binaries
      Includes
      Core
        Inc
        Src
          loader.c
          main.c
          secure_nsc.c
          stm32h5xx_hal_msp.c
          stm32h5xx_it.c
          syscalls.c
          systemem.c
          system_stm32h5xx_s.c
        Startup
      Drivers
      Debug
      STM32H573IIKXQ_FLASH.Id
      STM32H573IIKXQ_RAM.Id
      MX SecureAppli.ioc

Problems Tasks Console X Properties
CDT Build Console [SecureAppli_Secure]
arm-none-eabi-gcc ../Core/Src/loader.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DUSE_HAL_DRIVER -DSTM32H573XX -c
arm-none-eabi-gcc ../Core/Src/main.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DUSE_HAL_DRIVER -DSTM32H573XX -c -I
arm-none-eabi-gcc ../Core/Src/secure_nsc.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DUSE_HAL_DRIVER -DSTM32H573XX
arm-none-eabi-gcc ../Core/Src/stm32h5xx_hal_msp.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DUSE_HAL_DRIVER -DSTM3
arm-none-eabi-gcc ../Core/Src/stm32h5xx_it.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DUSE_HAL_DRIVER -DSTM32H573
arm-none-eabi-gcc ../Core/Src/syscalls.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DUSE_HAL_DRIVER -DSTM32H573XX -
arm-none-eabi-gcc ../Core/Src/systemem.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DUSE_HAL_DRIVER -DSTM32H573XX -c
arm-none-eabi-gcc ../Core/Src/system_stm32h5xx_s.c -mcpu=cortex-m33 -std=gnu11 -g3 -DDEBUG -DUSE_HAL_DRIVER -DSTM
arm-none-eabi-gcc -o "SecureAppli_Secure.elf" @"objects.list" -mcpu=cortex-m33 -T"C:\Training\STM32H5_Workshop\Ha
Finished building target: SecureAppli_Secure.elf

arm-none-eabi-size SecureAppli_Secure.elf
arm-none-eabi-objdump -h -S SecureAppli_Secure.elf > "SecureAppli_Secure.list"
arm-none-eabi-objcopy -O ihex SecureAppli_Secure.elf "SecureAppli_Secure.hex"
  text  data  bss  dec  hex filename
  6744   20  1580  8344  2098 SecureAppli_Secure.elf
Finished building: default.size.stdout

Finished building: SecureAppli_Secure.list
Finished building: SecureAppli_Secure.hex

"C:/Training/STM32H5_Workshop/Hands-on/Test_ST_iROT/SecureAppli/Secure/./ROT_Provisioning/postbuild.bat" "secure"
TPC success

14:02:08 Build Finished. 0 errors, 0 warnings. (took 7s.666ms)
  
```



# STiROT Hands-on

- ~~Step 0 : create an application with associated its bootpath~~
- ~~Step 1 : configure STiROT~~
- ~~Step 2 : develop your application~~
- Step 3 : provision STiROT config / DA access / Application
- Step 4 : secure update
- Step 5 : debug
- Step 6 : regression

## **Step 3 : provision STiROT config / DA access / Application**



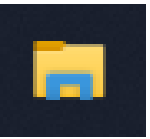
## STEP 3: Launch the provisioning of the board !

Domain > Training\_H5 > Hands-on > Test\_ST\_iROT > SecureAppli > ROT > Provisioning > ST\_iROT

| Name                     | Date modified      | Type               | Size  |
|--------------------------|--------------------|--------------------|-------|
| Image                    | 12/16/2022 4:04 PM | File folder        |       |
| Keys                     | 11/21/2022 1:47 PM | File folder        |       |
| ob_flash_programming.bat | 12/16/2022 3:53 PM | Windows Batch File | 3 KB  |
| obkey_provisioning.bat   | 11/21/2022 1:47 PM | Windows Batch File | 2 KB  |
| 1 provisioning.bat       | 12/16/2022 1:39 PM | Windows Batch File | 11 KB |
| update_and_setup.bat     | 12/16/2022 1:40 PM | Windows Batch File | 6 KB  |
| update_console_setup.bat | 12/16/2022 1:40 PM | Windows Batch File | 3 KB  |

Hands-on\Test\_ST\_iROT\SecureAppli\ROT\Provisioning\STiROT

provisioning.bat



# STEP 3: Provisioning script

```
=====  
===== Provisioning of STiRoT boot path  
===== Application selected through env.bat:  
===== C:\Training\STM32H5_Workshop\Hands-on\Test_ST_iRoT\SecureAppli\RoT_Provisioning\...\br/>===== Product state must be Open. Execute \RoT_Provisioning\DA\regression.bat if not the case.  
=====
```

```
    * ob_flash_programming script update ...  
      Option bytes successfully updated according to STiRoT_Config.xml  
      (see "update_ob_setup.log" for details)
```

```
Step 2 : Images generation  
    * Data generation (if Data image is enabled):  
      Select STiRoT_Data_Image.xml(Default path is \RoT_Provisioning\STiRoT\Image\STiRoT_Data_Image.xml)  
      Generate the data_enc_sign.hex image  
      Press any key to continue...
```

```
Step 3 : Provisioning  
    * BOOT0 pin should be disconnected from VDD:  
      (STM32H573I-DK: set SW1 to position 0)  
      Press any key to continue...
```

```
    * Programming the option bytes and flashing the images ...  
      Successful option bytes programming and images flashing  
      (see "ob_flash_programming.log" for details)
```

```
    * Define product state value  
      [ OPEN | PROVISIONED | TZ-CLOSED | CLOSED | LOCKED ] : CLOSED
```

```
    * Setting the product state PROVISIONING
```

```
    * Provisioning the .obk files ...  
      Successful obk provisioning  
      (see "obkey_provisioning.log" for details)
```

```
    * Setting the final product state CLOSED
```

```
=====  
===== The board is correctly configured.  
===== Power off/on the board to start the application.  
=====
```

STiRoT\_Config already up-to-date

User data image signature up-to-date

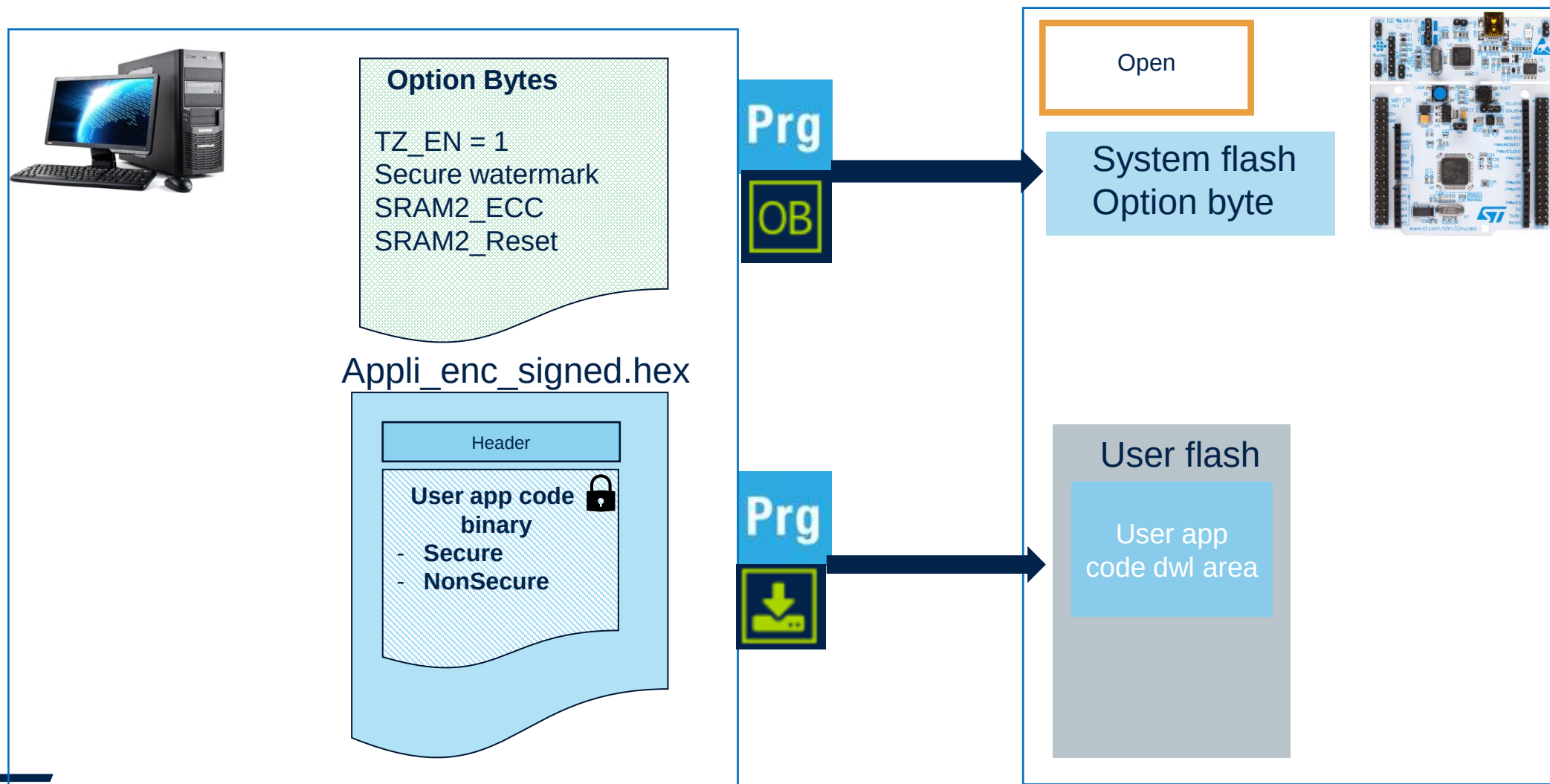
Everything is ready. Start  
provisioning the board

Start flashing image and setting  
option bytes configuration

Select final state : **CLOSED**  
Switch to PROVISIONING state  
Provision 3 obk files: **DA**,  
**STiRoT\_Config**, **STiRoT\_Data**  
Then set selected final state

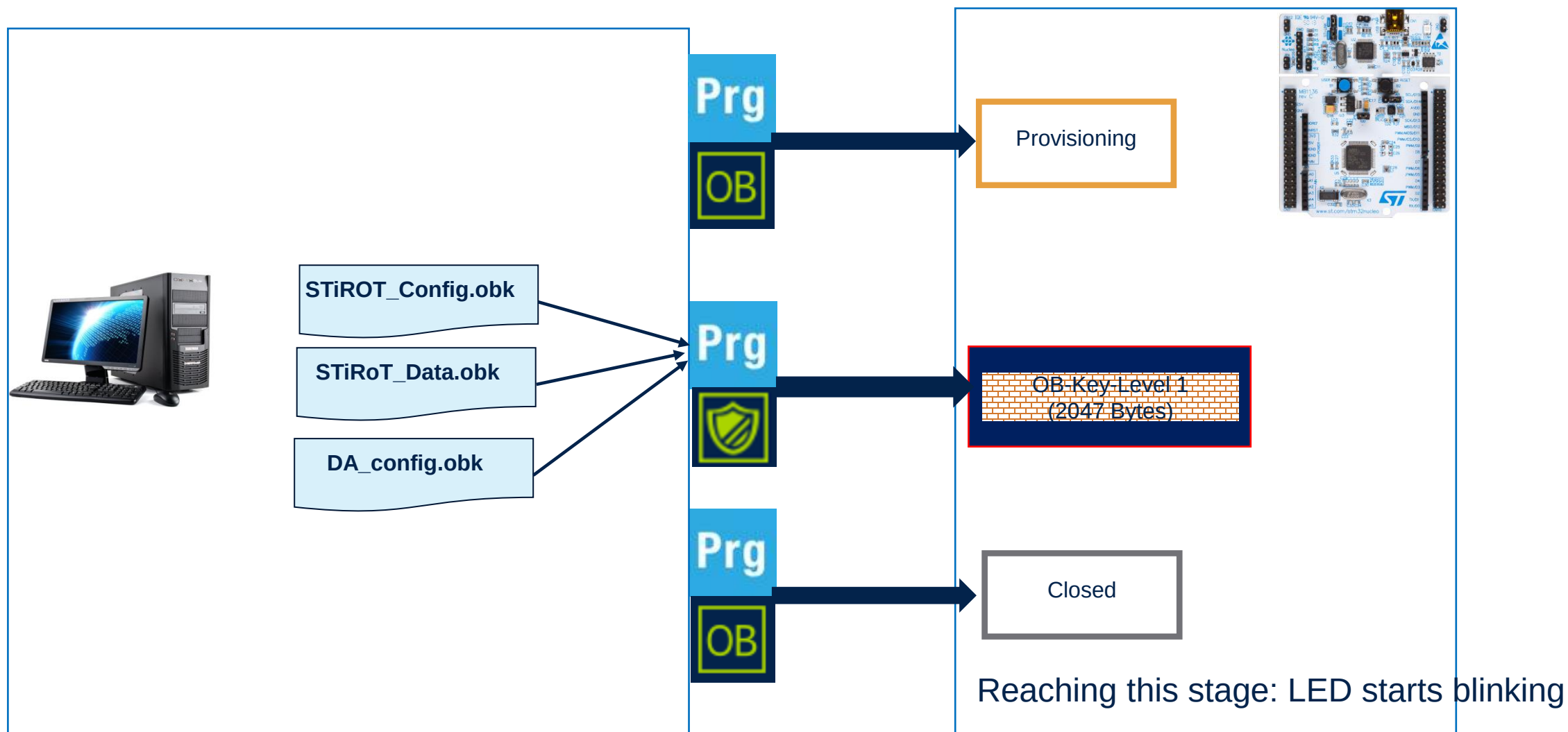


# STEP 3: STiROT Provisioning phase 1





# STEP 3: STiROT Provisioning phase 2

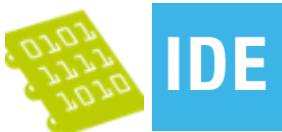




# STiROT Hands-on

- ~~Step 0 : create an application with associated its bootpath~~
- ~~Step 1 : configure STiROT~~
- ~~Step 2 : develop your application~~
- ~~Step 3 : provision STiROT config / DA access / Application~~
- Step 4 : secure update
- Step 5 : debug
- Step 6 : regression

## Step 4 : secure update



# STEP 4: LED Blinking ! Modify your application....

workspace\_1.13.1 - SecureAppli\_Secure/Core/Src/main.c - STM32CubeIDE

File Edit Source Refactor Navigate Search Project Run Window Help myST

Project Explorer

- SecureAppli
  - Binary
  - Drivers
  - ROT\_Provisioning
  - Secure\_nsclib
  - SecureAppli\_Secure (in Secure)
    - Binaries
    - Includes
    - Core
      - Inc
      - Src
        - loader.c
        - main.c
        - secure\_nsc.c
        - stm32h5xx\_hal\_msp.c
        - stm32h5xx\_it.c
        - syscalls.c
        - system.c
        - system\_stm32h5xx\_s.c
      - Startup
      - Drivers
      - Debug
      - STM32H573IIXQ\_FLASH.Id
      - STM32H573IIXQ\_RAM.Id
    - SecureAppli.ioc

main.c

```
100  /* Infinite loop */
101  /* USER CODE BEGIN WHILE */
102  while (1)
103  {
104      /* USER CODE END WHILE */
105
106      /* USER CODE BEGIN 3 */
107      HAL_GPIO_TogglePin(GPIOI,GPIO_PIN_9);
108      HAL_Delay(100);
109      if (HAL_GPIO_ReadPin(GPIOC,GPIO_PIN_13) == 0x1)
110      {
111          /* USER CODE BEGIN 4 */
112          /* USER CODE END 4 */
113      }
114      /* USER CODE BEGIN 5 */
115      /* USER CODE END 5 */
116  }
117  /**
118   * @brief GTZC_S Initialization Function
119   * @param None
120   * @retval None
121   */
122  static void MX_GTZC_S_Init(void)
123  {
124      /* USER CODE BEGIN GTZC_S_Init 0 */
125
126      /* USER CODE END GTZC_S_Init 0 */
127
128      MPCBB_ConfigTypeDef MPCBB_Area_Desc = {0};
129
130      /* USER CODE BEGIN GTZC_S_Init 1 */
131  }
```

1

HAL\_Delay(100);

2

Problems Tasks Console Properties

CDT Build Console [SecureAppli\_Secure]

"C:/Training/STM32H5\_Workshop/Hands-on/Test\_ST\_iROT/SecureAppli/Secure/./ROT\_Provisioning/postbuild.bat" "secure"

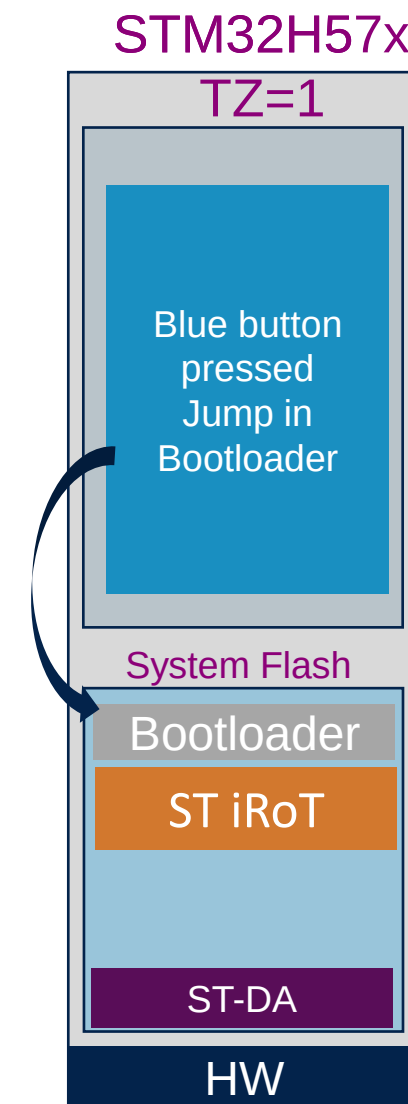
TPC success

14:23:14 Build Finished. 0 errors, 0 warnings. (took 4s.531ms)

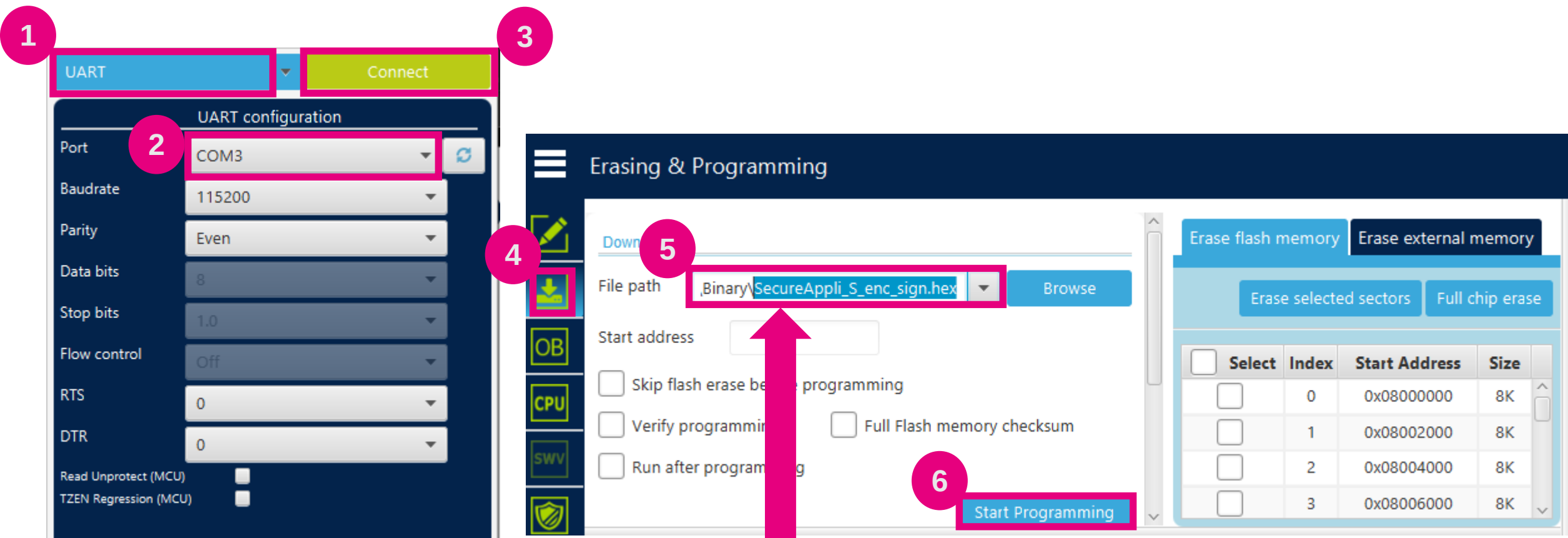


# Target

- When we press the button, the application will jump in the embedded bootloader.
- You can then connect to bootloader thanks to USART or USB



# STEP 4: Press Blue button to enter in bootloader



1. UART

2. COM3

3. Connect

4. Download

5. File path: .Binary\SecureAppli\_S\_enc\_sign.hex

6. Start Programming

UART configuration

Port: COM3

Baudrate: 115200

Parity: Even

Data bits: 8

Stop bits: 1.0

Flow control: Off

RTS: 0

DTR: 0

Read Unprotect (MCU): ☐

TZEN Regression (MCU): ☐

Erasing & Programming

Download

File path: .Binary\SecureAppli\_S\_enc\_sign.hex

Browse

Start address

☐ Skip flash erase before programming

☐ Verify programming

☐ Run after programming

☐ Full Flash memory checksum

Start Programming

Erase flash memory

Erase external memory

Erase selected sectors

Full chip erase

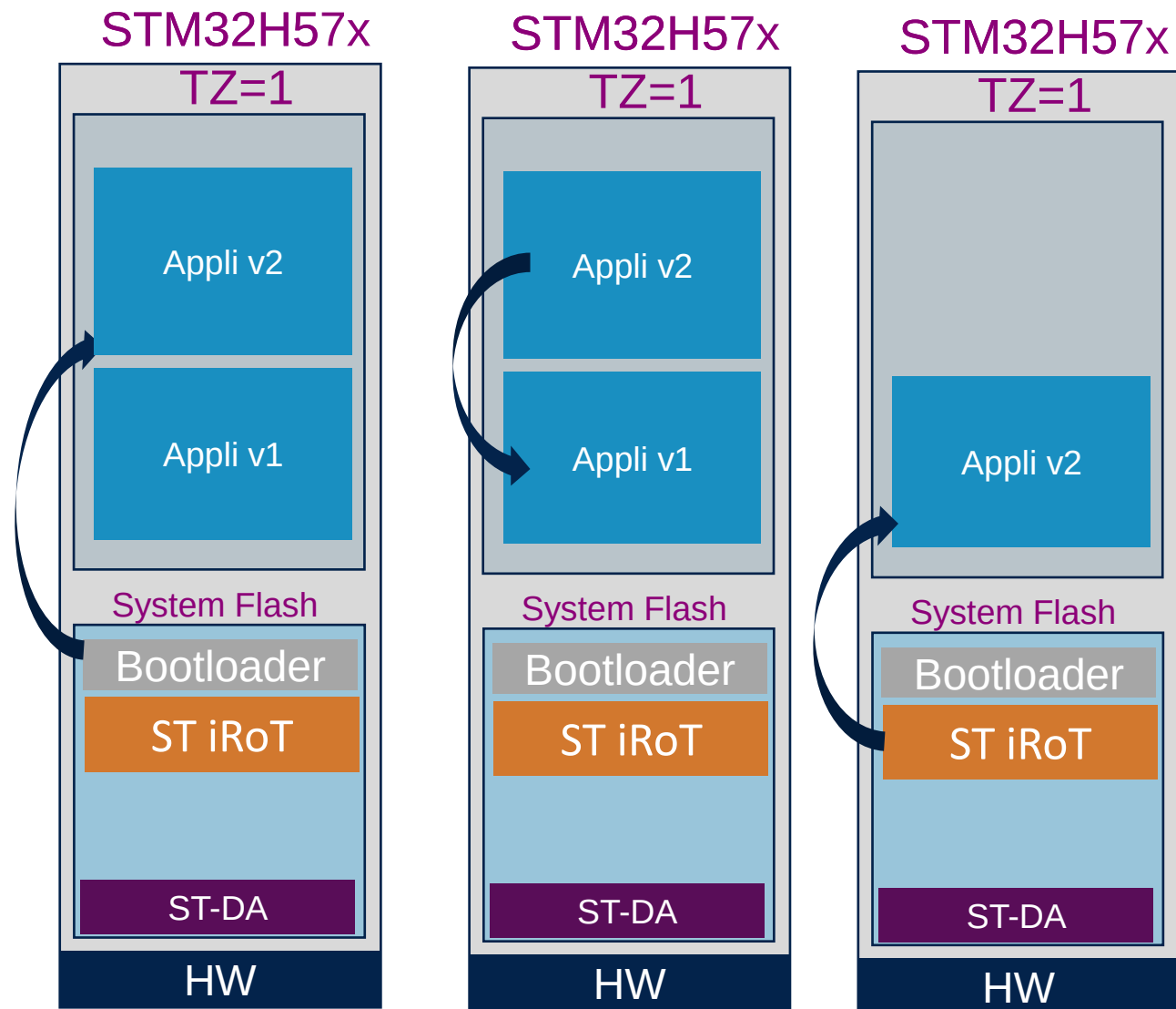
| Select                   | Index | Start Address | Size |
|--------------------------|-------|---------------|------|
| <input type="checkbox"/> | 0     | 0x08000000    | 8K   |
| <input type="checkbox"/> | 1     | 0x08002000    | 8K   |
| <input type="checkbox"/> | 2     | 0x08004000    | 8K   |
| <input type="checkbox"/> | 3     | 0x08006000    | 8K   |

C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT\SecureAppli\Binary\SecureAppli\_S\_enc\_sign.hex



## STEP 4: Secure update the application

- Thanks bootloader we download a new version of the application
- At next reset, STiROT detect a new version, check authenticity and integrity. If ok install.
- Then it jump on the new firmware !
- So press the reset button on your target...

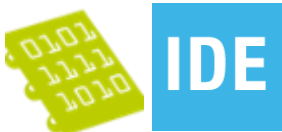




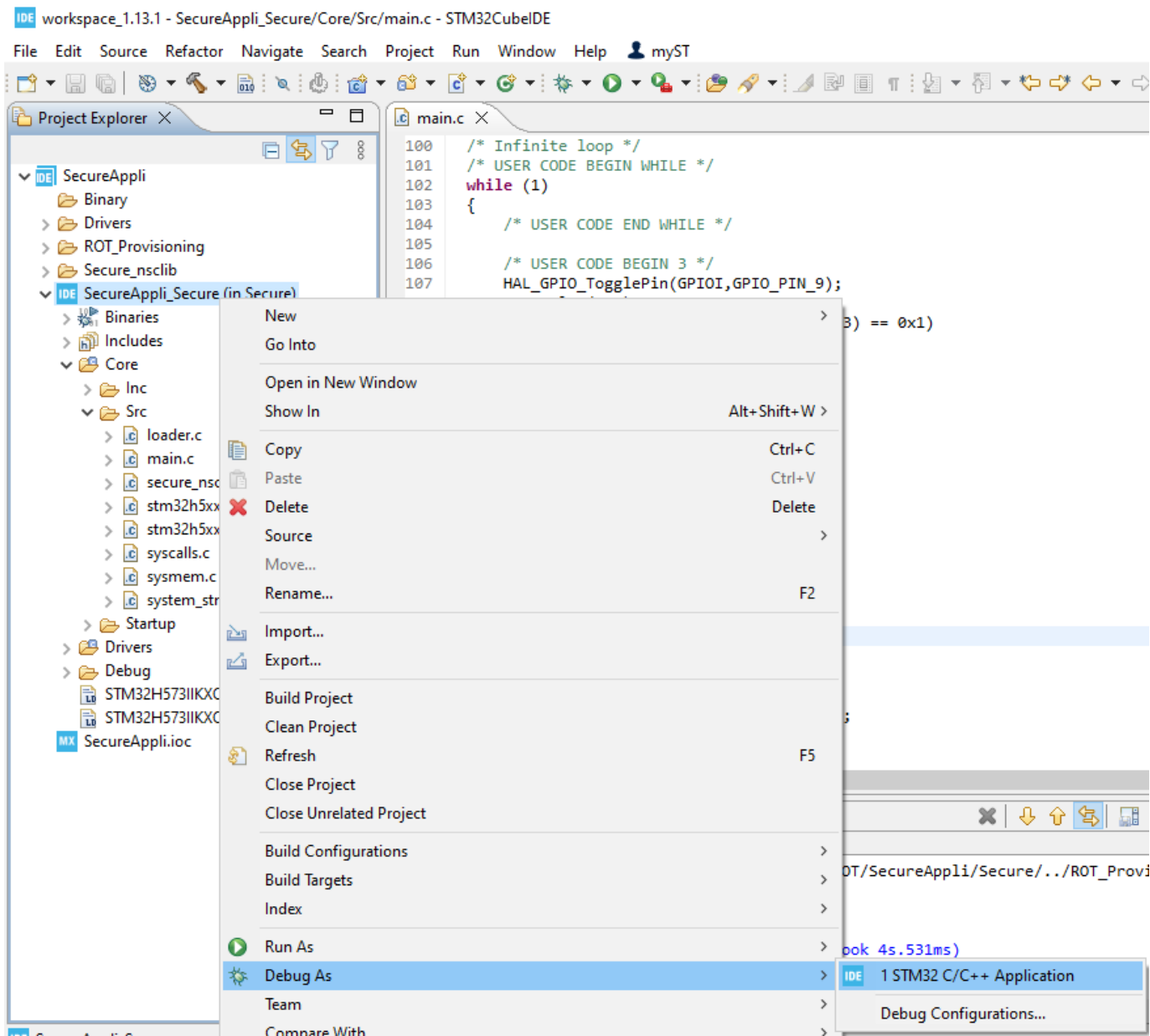
# STiROT Hands-on

- ~~Step 0 : create an application with associated its bootpath~~
- ~~Step 1 : configure STiROT~~
- ~~Step 2 : develop your application~~
- ~~Step 3 : provision STiROT config / DA access / Application~~
- ~~Step 4 : secure update~~
- Step 5 : debug
- Step 6 : regression

## Step 5 : debug



# STEP 5: Launch a debug session.



# STEP 5: configure debug session.

IDE Edit Configuration

Edit launch configuration properties



1 SecureAppli Secure

Main **Debugger** Startup Source Common

GDB Connection Settings

☒ Autostart local GDB server Host name or IP address localhost

☐ Connect to remote GDB server Port number 61234

Debug probe ST-LINK (ST-LINK GDB server)

GDB Server Command Line Options

Show Command Line

Interface

☒ SWD ☐ JTAG

☐ ST-LINK S/N Scan

Frequency (kHz): Auto

Access port: 1 - Cortex-M33

2 Reset behaviour

Type: Software system reset

3 Debug Authentication

☒ Enable

Key path: C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT\SecureAppli\ROT\_Provisioning\DA\Keys\key\_3\_leaf.pem Browse...

Certificate path: C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT\SecureAppli\ROT\_Provisioning\DA\Certificates\cert\_leaf\_chain.b64 Browse...

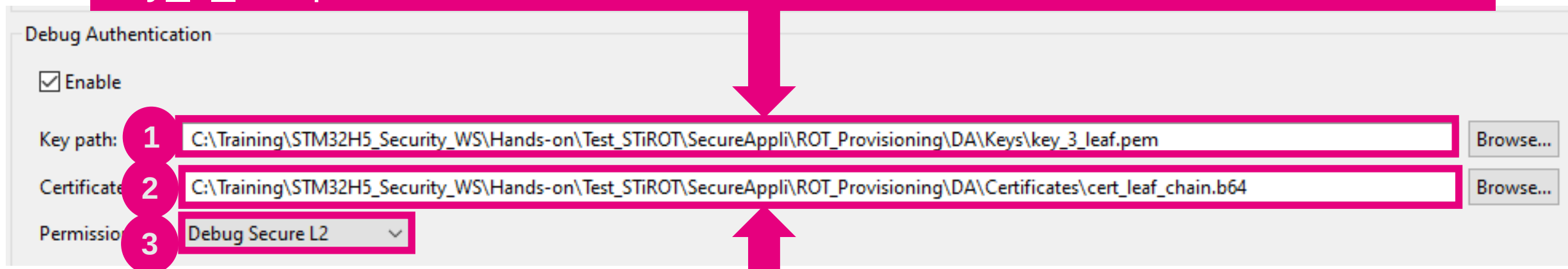
Permissions: Debug Secure L2

Revert Apply

OK Cancel

## STEP 5: configure DA access.

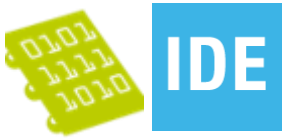
C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT\SecureAppli\ROT\_Provisioning\DA\Keys\key\_3\_leaf.pem



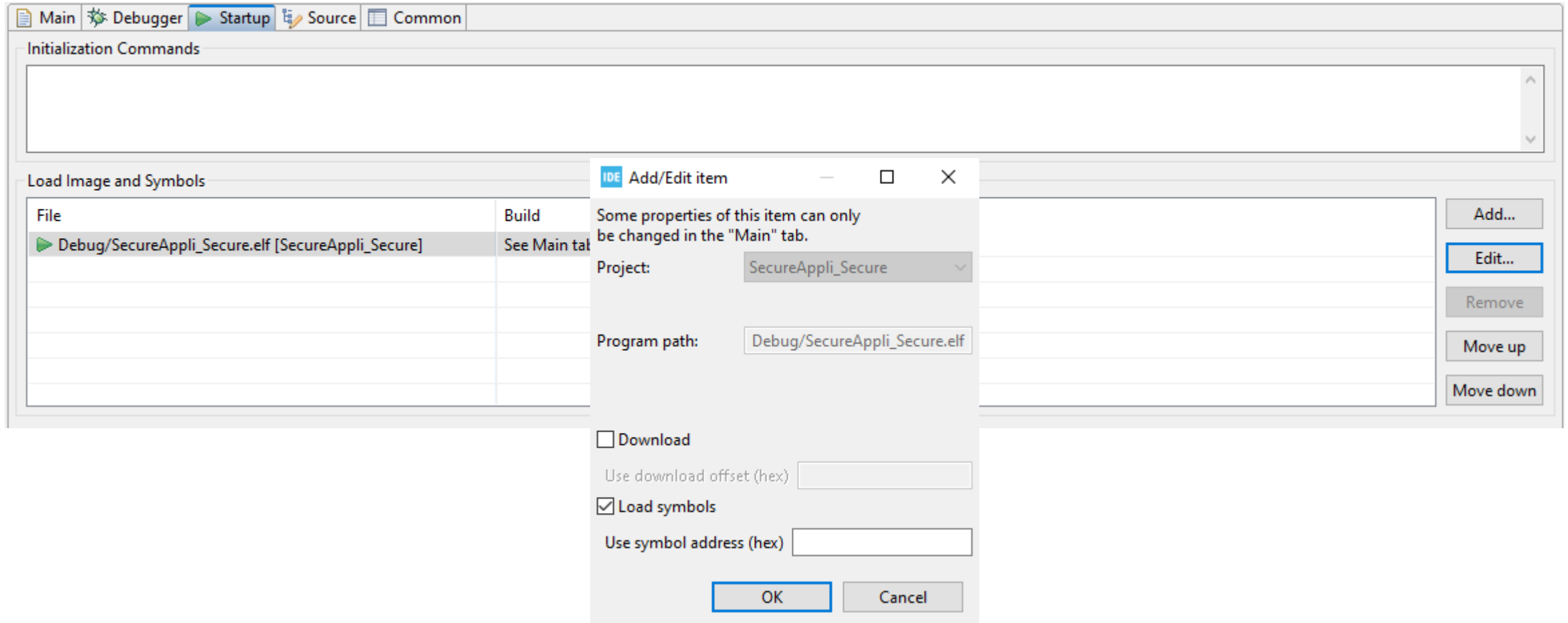
The screenshot shows the 'Debug Authentication' dialog box. It has a 'Key path' field, a 'Certificate' field, and a 'Permission' dropdown. Three numbered callouts are present: 1 points to the 'Key path' field, 2 points to the 'Certificate' field, and 3 points to the 'Permission' dropdown. A large pink arrow points from the top text box to the 'Key path' field, and another large pink arrow points from the bottom text box to the 'Certificate' field.

| Field       | Value                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------|
| Key path    | C:\Training\STM32H5_Security_WS\Hands-on\Test_STiROT\SecureAppli\ROT_Provisioning\DA\Keys\key_3_leaf.pem              |
| Certificate | C:\Training\STM32H5_Security_WS\Hands-on\Test_STiROT\SecureAppli\ROT_Provisioning\DA\Certificates\cert_leaf_chain.b64 |
| Permission  | Debug Secure L2                                                                                                       |

C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT\SecureAppli\ROT\_Provisioning\DA\Certificates\cert\_leaf\_chain.b64

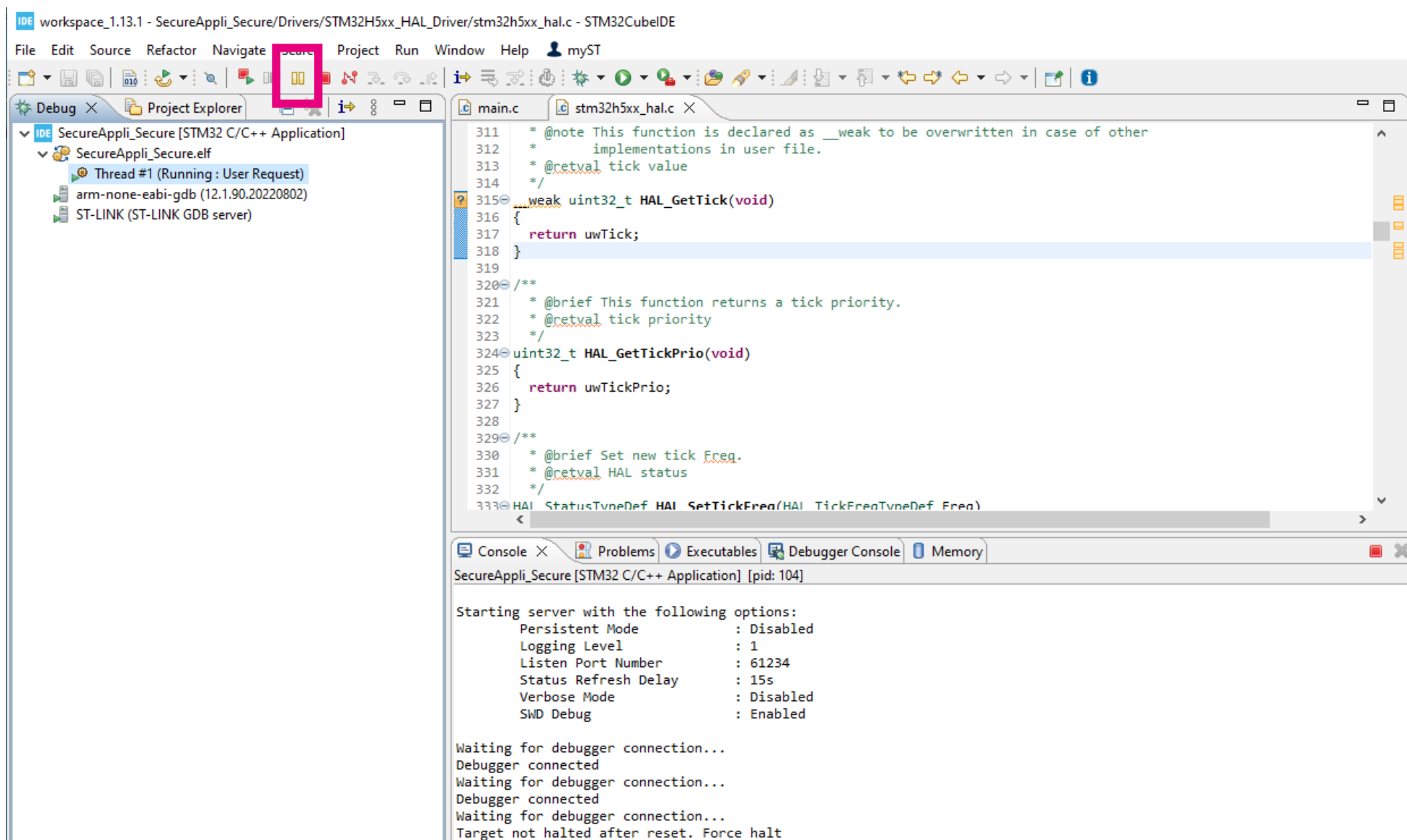


# STEP 5: remove the download



Select OK and then launch Debug

# STEP 5: Stop execution..

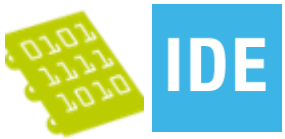


The screenshot shows the STM32CubeIDE interface. The main editor displays the `stm32h5xx_hal.c` file, specifically the `HAL_GetTick` function. The function is highlighted in blue, and the `return uwTick;` line is also highlighted. The Project Explorer on the left shows the project structure, including the `SecureAppli_Secure` project and the `Thread #1 (Running : User Request)` task. The bottom panel shows the Debugger Console with the following output:

```
SecureAppli_Secure [STM32 C/C++ Application] [pid: 104]

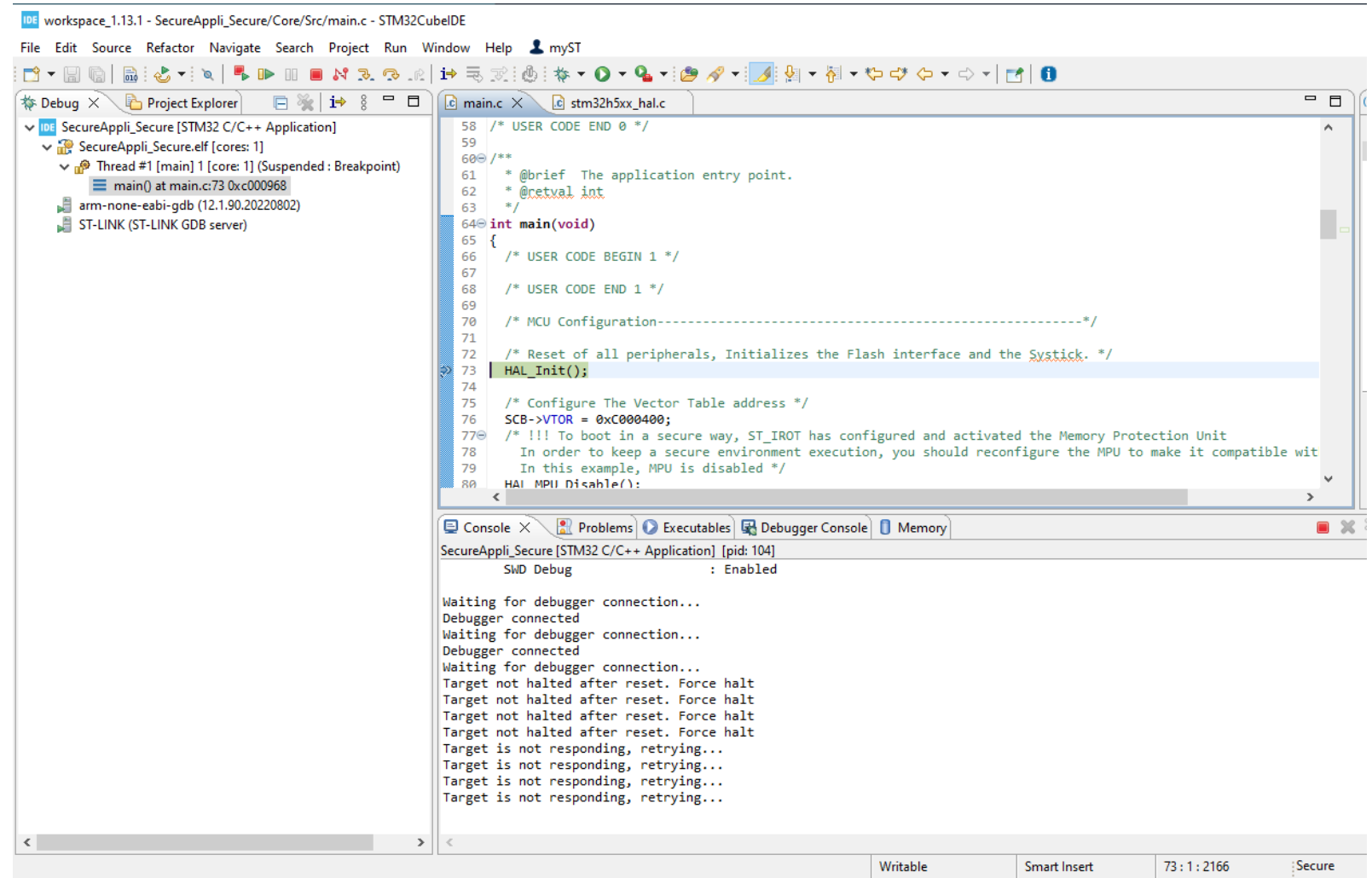
Starting server with the following options:
  Persistent Mode      : Disabled
  Logging Level       : 1
  Listen Port Number  : 61234
  Status Refresh Delay : 15s
  Verbose Mode        : Disabled
  SWD Debug           : Enabled

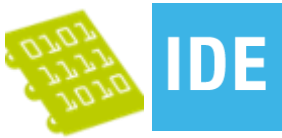
Waiting for debugger connection...
Debugger connected
Waiting for debugger connection...
Debugger connected
Waiting for debugger connection...
Target not halted after reset. Force halt
```



# STEP 5: Set break point on HAL\_Init and reset

- Reset should be done with reset button on the board
- Debugger stops on breakpoints as usual





# STEP 5: Stop debugging session

IDE workspace\_1.13.1 - SecureAppli\_Sec /Src/main.c - STM32CubeIDE

File Edit Source Refactor Naviga Project Run Window Help myST

Debug X Project Explorer

SecureAppli\_Secure [STM32 C/C++ Application]  
SecureAppli\_Secure.elf [cores: 1]  
Thread #1 [main] 1 [core: 1] (Suspended : Breakpoint)  
main() at main.c:73 0xc000968  
arm-none-eabi-gdb (12.1.90.20220802)  
ST-LINK (ST-LINK GDB server)

main.c x stm32h5xx\_hal.c

```
58 /* USER CODE END 0 */
59
60 /**
61  * @brief The application entry point.
62  * @retval int
63  */
64 int main(void)
65 {
66     /* USER CODE BEGIN 1 */
67
68     /* USER CODE END 1 */
69
70     /* MCU Configuration-----*/
71
72     /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
73     HAL_Init();
74
75     /* Configure The Vector Table address */
76     SCB->VTOR = 0xC000400;
77     /* !!! To boot in a secure way, ST_IROT has configured and activated the Memory Protection Unit
78      In order to keep a secure environment execution, you should reconfigure the MPU to make it compatible wit
79      In this example, MPU is disabled */
80     HAL_MPU_Disable();
```

Console X Problems Executables Debugger Console Memory

SecureAppli\_Secure [STM32 C/C++ Application] [pid: 104]  
SWD Debug : Enabled

Waiting for debugger connection...  
Debugger connected  
Waiting for debugger connection...  
Debugger connected  
Waiting for debugger connection...  
Target not halted after reset. Force halt  
Target not halted after reset. Force halt  
Target not halted after reset. Force halt

# STEP 5: Close the debug ...

STM32CubeProgrammer

Secure programming

2

RDP REG SFI/SFIx PROV **DA** SSP

Key File Path

Select File  Browse

Certificate File Path

1  Browse Continue

 This interface supports only STM32H5!

Discover

| name              | value                                        |
|-------------------|----------------------------------------------|
| Locking Mechanism | Certificate                                  |
| Soc ID            | 0x003A001A 0x3332510F 0x38363236 0x000000... |
| Life Cycle        | ST_LIFECYCLE_CLOSED                          |
| Device ID         | 0x484                                        |

3

Close Debug

Used to lock device once the debug has been opened through Debug Authentication process. Applicable only when the feature is available.

Message

Target successfully locked.

4

OK



# STiROT Hands-on

- ~~Step 0 : create an application with associated its bootpath~~
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- ~~Step 4 : secure update~~
- ~~Step 5 : debug~~
- Step 6 : regression

# Step 6 : Regression

# STEP 6: Let's do a regression...

STM32CubeProgrammer

STM32CubeProgrammer

Data Information Notice

## Secure programming

RDP REG SFI/SFIx PROV **DA** SSP

Key File Path

Select File  Browse

Certificate File Path

Select File  Browse Continue

*This interface supports only STM32H5!*

**1** Discover

| name              | value                                        |
|-------------------|----------------------------------------------|
| Locking Mechanism | Certificate                                  |
| Soc ID            | 0x003A001A 0x3332510F 0x38363236 0x000000... |
| Life Cycle        | ST_LIFECYCLE_CLOSED                          |
| Device ID         | 0x484                                        |

Close Debug

Used to lock device once the debug has been opened through Debug Authentication process. Applicable only when the feature is available.

## STEP 6: Let's do a regression...

C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT\SecureAppli\ROT\_Provisioning\DA\Keys\key\_3\_leaf.pem

Key File Path

1 Select File C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT\SecureAppli\ROT\_Provisioning\DA\ Browse

Certificate File Path

2 Select File \_WS\Hands-on\Test\_STiROT\SecureAppli\ROT\_Provisioning\DA\Certificates\cert\_leaf\_chain.b64 Browse 3 Continue

C:\Training\STM32H5\_Security\_WS\Hands-on\Test\_STiROT\SecureAppli\ROT\_Provisioning\DA\Certificates\cert\_leaf\_chain.b64

# STEP 6: Select a full regression

Permissions

| Permission                          | Select                           |
|-------------------------------------|----------------------------------|
| Non-Secure Intrusive Debug (Level1) | <input type="radio"/>            |
| Non-Secure Intrusive Debug (Level2) | <input type="radio"/>            |
| Non-Secure Intrusive Debug (Level3) | <input type="radio"/>            |
| Secure Intrusive Debug (Level1)     | <input type="radio"/>            |
| Secure Intrusive Debug (Level2)     | <input type="radio"/>            |
| Secure Intrusive Debug (Level3)     | <input type="radio"/>            |
| Partial Regression                  | <input type="radio"/>            |
| Full Regression                     | <input checked="" type="radio"/> |

1

2

Execute

3

OK

Message

Debug Authentication Success

# STEP 6: Check the product state

1

OB

Option bytes

▼ Product state

| Name          | Value                        | Description                                                                                                                                                                                                                                                                                       |
|---------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PRODUCT_STATE | <div>2</div> <div>ED ▼</div> | Life state code.<br>ED : Open<br>17 : Provisioning, Debug partially opened (only non-secure)<br>2E : iRoT-provisioned, Debug partially opened (only non-secure)<br>C6 : TZ-Closed, Debug partially opened (only non-secure)<br>72 : Closed, Debug disabled, regression is possible<br>5C : Locked |



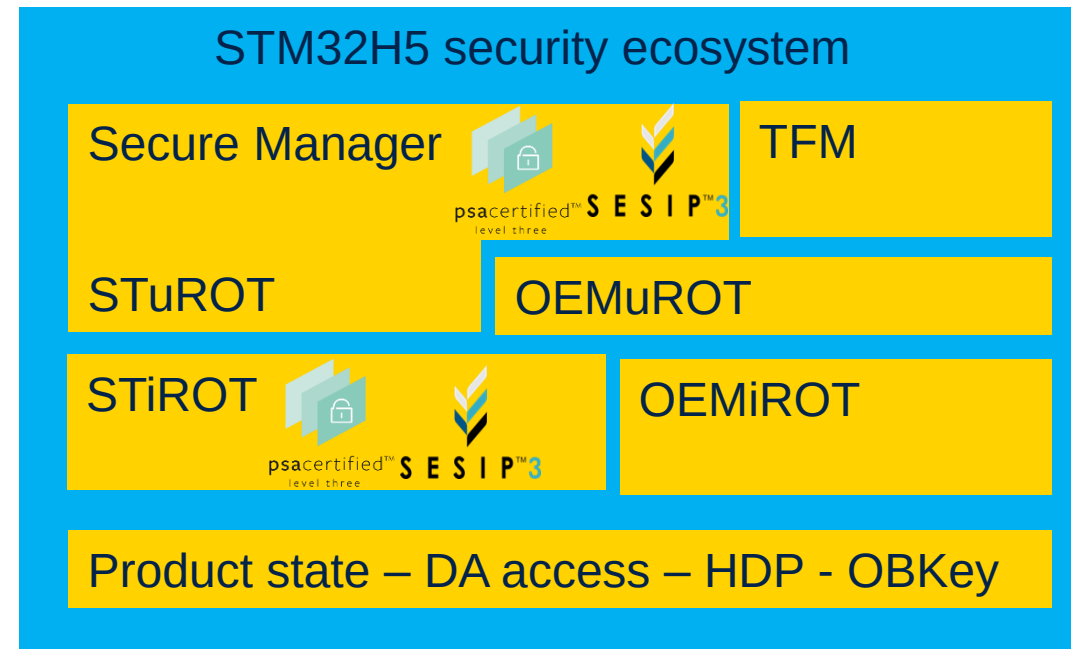
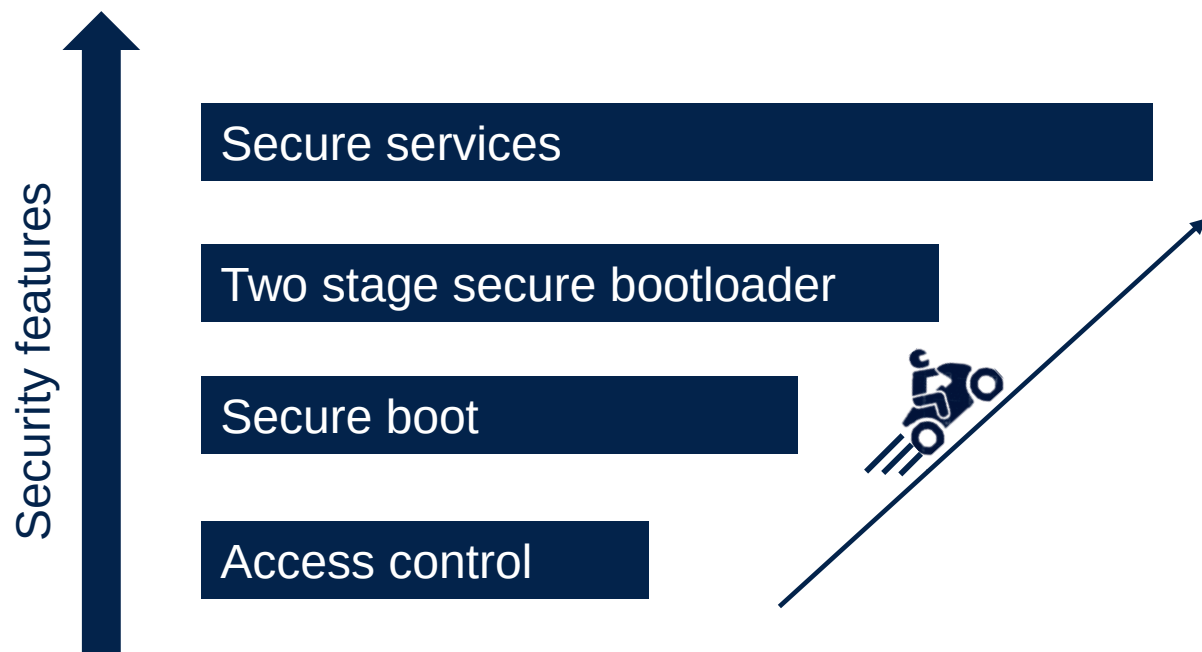
# STiROT Hands-on

- We have seen
  - How to STiROT is integrated in the STM32 ecosystem
  - How to create an application with CubeMX and STiROT Boot Path
  - How to do flash the target thanks to the delivered provisioning script
  - How to perform a secure update
  - How to open the debug of a closed device and debug the installed application
  - How to close this debug and perform full regression

# Conclusion



# Scalable security





# Conclusion

- The STM32H5 introduces new security mechanism :
  - A new temporal isolation level
  - A secure storage coupled with temporal isolation
  - New product states
  - The Debug Authentication
  - An embedded secure boot with secure firmware update capability
  - Code example of two stage bootloader open sources
  - Crypto accelerator side channel attack resistant.
- All those features allow to achieve SESIP/PSA level 3 certification.
- You can select and build your own security adapted to your needs !

# STM32H5 security resources

- [https://www.st.com/content/st\\_com/en/wiki/wiki-portal.html](https://www.st.com/content/st_com/en/wiki/wiki-portal.html)  
<https://wiki.st.com/stm32mcu/wiki/Category:Security>
- [https://www.st.com/content/st\\_com/en/support/learning/stm32-education/stm32-online-training/stm32h5-online-training.html](https://www.st.com/content/st_com/en/support/learning/stm32-education/stm32-online-training/stm32h5-online-training.html)

## Security

|                                  |                                                                                       |                                                                                       |
|----------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Security Architecture Overview   |    |    |
| Enhanced Key Storage (KEYSTOR)   |    |    |
| Life cycle                       |    |    |
| Debug Authentication             |    |    |
| Secure Manager (SECMGR)          |   |   |
| Root Security Services (SFI)     |  |  |
| Hash and Random Number Generator |  |  |
| Security Certification           |  |  |
| Crypto                           |  |  |
| Symmetric Crypto                 |  |  |
| Asymmetric crypto                |  |  |
| Crypto Lib                       |  |  |

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